

Teacher Guide — Electrostatics (charges, Coulomb's law, electric field, flux, Gauss's law)

Physics | Grade 11-12

Period 1

ENTRY TICKET

You rub a dry balloon on your hair. After rubbing, the balloon sticks to the wall. Circle which statement you think is true, and briefly say why (one sentence):

- A) Rubbing the balloon created new electric charge on the balloon and on the wall.
 - B) Rubbing moved existing charge so the balloon gained negative charge and some other object lost the same amount of negative charge.
 - C) The balloon became charged only because your hair produced charge from nowhere.
- (Write A, B, or C and one short reason.)

Expected response: Most correct response: B — rubbing transfers electrons so the balloon gains negative charge and something else loses the same amount; no net charge is created. A common wrong response: A or C, written with reasons like 'rubbing makes charge' or 'hair makes charge'.

Teacher Script

0-5 min: Collect responses and surface the key belief

Say: "Turn your paper over so I can see your letter and one-line reason. Hold it up when I call your row." Walk to the first row, pick two student answers to read aloud (one that picked B and one that picked A/C). Then say: "Thanks — we will check which view fits the evidence. Keep your paper — we may return to it."

Board: Write: "Entry ticket: A / B / C" and tally two example answers under each as you read them aloud.

- Anticipate: What if a balloon is neutral after rubbing? (Students may ask if rubbing always charges.)
- Anticipate: How can we tell which object lost electrons? (Students want to know direction of transfer.)
- Anticipate: Does the wall get charged if it is plaster or painted? (Asking about material differences.)

5-15 min: Introduce electric charge at the atomic scale

Say: "At the atomic level, some particles — electrons — carry negative charge and can move between materials. Protons stay fixed in atomic nuclei. When two materials touch and rub, electrons can move from atoms in one material to atoms in the other. That is a transfer of existing electrons — not creation of new charge." Point to the diagram of an atom and label electron and nucleus. Then ask: "Which particle moves during rubbing, electron or proton?" Wait for the answer and confirm: "Electron."

Then say: "So when a balloon picks up extra electrons, it becomes negatively charged by gain of electrons; the object that lost those electrons becomes positively charged by the same amount."

Board: Draw a simple atom: small labeled nucleus and orbiting electrons; label 'electron (negative)' and 'proton in nucleus (positive)'. Underline: 'electrons can move between materials; protons do not'.

- Anticipate: How many electrons move? (Students wonder about magnitude — answer: varies; we will not calculate charge yet.)
- Anticipate: Why don't protons move? (Answer: protons are bound in nuclei and much more tightly held.)
- Anticipate: What makes electrons move from one material to another? (Lead into next beat: different materials hold electrons with different strengths.)

15-27 min: Teacher demonstration: rubbing and showing transfer

Perform demonstration: Use a nylon rod and fur, and an electroscope (or a small aluminium foil leaf electroscope). Steps aloud as you do them:

- 1) "Show the uncharged electroscope: leaves hang close together." Place the rod nearby without touching — note no effect.
- 2) "Now rub the nylon rod with the fur for 20 seconds." Rub the rod visibly while students watch.
- 3) "Touch the electroscope knob briefly with the rubbed rod." Touch and then step back and point: "The leaves separate — that shows charge was transferred to the electroscope."
- 4) "Tap the electroscope to ground it and then touch again with the rod to show charging by contact repeats." Ask: "Did we create net charge inside the room?" Wait for answers and guide to: "No — we moved electrons from one object to another; the total charge in the closed system stayed the same."

Board: Write the sequence of actions as short bullets: 'uncharged electroscope -> rub rod -> touch electroscope -> leaves separate'. Draw a simple electroscope sketch with leaves apart after charging.

- Anticipate: Where did the electrons come from? (Answer: from the fur or the rod depending on which material loses electrons.)
- Anticipate: If I ground the electroscope, what happens? (Answer: ground provides a path for electrons to leave or enter; next step shows leaves collapse.)
- Anticipate: Does the electroscope show positive vs negative? (Answer: it shows net charge magnitude via leaf separation; sign depends on which object was touched.)

27-33 min: Mini-lab: short guided pairing to test conductors vs insulators

Say: "In pairs, take a plastic comb and a small metal paperclip. Rub the comb on your hair for 15 seconds, then bring it near the paperclip (not touching). Observe what happens. Then touch the comb briefly to one end of the paperclip and observe. 3 minutes to do this; I will circulate." After 3 minutes call them back and ask two pairs to report: what happened near the paperclip, and what happened after touching. Key point to elicit: comb (an insulator) can hold localized charge and attract the paperclip; metal paperclip conducts and any charge given to it flows and may be shared or lost to teacher's grounded table.

Board: Write instructions briefly: 'Combine: comb + hair -> bring near paperclip (no touch) -> touch paperclip'.

Underneath write: 'Observe: attraction? leaves? charge retention?'.

- Anticipate: Why did the paperclip stick sometimes but not always? (Answer: induced charge or grounding; also humidity and contact matter.)
- Anticipate: Can the metal become charged? (Answer: yes, but it conducts so charge spreads or leaks to ground.)
- Anticipate: If the comb is an insulator why does it attract the metal? (Answer: comb holds

charge locally; it induces opposite charge in metal causing attraction.)

33-37 min: Guided practice: explain rubbing as transfer, not creation

Project or write this prompt and call on volunteers: "Explain in 30 seconds why rubbing does not create net charge. Use the terms electrons, transfer, and conservation." Call on 3 students. For each response, give a one-sentence corrective comment: e.g. 'Good — you named electrons and transfer; add that the other object loses the same number of electrons so net charge is unchanged.'

Board: Write the three key words students must include: 'electrons', 'transfer', 'conserved (net unchanged)'.

- Anticipate: What does 'net' charge mean? (Answer: sum of positive and negative — if electrons are moved within the system total doesn't change.)
- Anticipate: Can rubbing create equal numbers of positive and negative charges? (Guide to: charges are transferred as electrons; you do not create new positive or negative out of nothing.)

37-40 min: Check for understanding and link forward

Say: "Quick quiz: True or false — An insulator cannot become charged. Hold up T or F." Scan responses; then say: "Correct answer: False. Insulators can acquire and hold charge locally. Conductors let charges move and may leak to ground. Next period we will use these ideas to explain why charge comes in packets (quantisation) and to derive the mathematical form of Coulomb's law." Collect one final verbal summary from the class: 'Charge is electrons moving; rubbing transfers electrons; total charge conserved.'

Board: Write two short bullets linking forward: 'Next: quantisation of charge' and 'Then: Coulomb's law'.

- Anticipate: Will we calculate forces next time? (Answer: yes — Coulomb's law gives magnitude and vector later.)
- Anticipate: Do we ever get neutral objects after contact? (Answer: yes, if net transfer cancels; we will discuss how to measure net charge later.)

Blackboard Notes

- Charge, Conductors, and Conservation
- Key words to include
- insulator: A material that offers high resistance to the passage of electricity and in which charge placed remains localized rather than spreading.
- At atomic scale: electrons move between materials; protons remain in nuclei.
- Rubbing transfers electrons from one object to another — it does NOT create net charge.
- Conductors: charges are comparatively free to move; insulators: charges stay localized.
- Conservation of charge: in an isolated system the total electric charge is conserved; charge may be redistributed but the net charge remains constant.
- Next: quantisation of charge -> then Coulomb's law (magnitude & vector)
- Diagram: Simple atom: small circle labelled 'nucleus (protons + neutrons)' and one or two electrons drawn and labelled 'electron (negative)'.
- Diagram: Electroscope sketch: knob and two leaves, with leaves apart after contact; arrow pointing from rubbed rod to knob showing electron transfer.
- Diagram: Comb near paperclip: comb with '-' signs on its surface and induced '+' signs on the nearer end of the paperclip (no formula, just signs to show transfer/induction).

Activities

Rubbing Materials and Tracking Electrons (12 min, experiment)

Materials: 2 balloons (one per pair), small wool or cotton cloth (one per pair), plastic comb or plastic pen (one per pair), small metal key or paperclip (one per pair), two 10 cm squares of aluminum foil (one pair share), piece of sticky tape, marker, worksheet slip (teacher-printed, one per pair) — optional

- 1. Arrange students in pairs. Give each pair the listed materials. Say: “We will rub objects and watch where charges move. Our goal: show whether rubbing creates net charge or only moves electrons between materials.” Start a visible 12-minute timer.
- 2. Tell students to follow the student instructions. While they work, move between pairs. For the first 6 minutes watch that each pair performs the two rub tests in order (balloon–wool and comb–wool) and that they label which object was rubbed and which was not. Do not answer interpretive questions; instead ask, “What did you observe? Which object moved electrons according to your test?”
- 3. After 6 minutes give the prompt: “Now test whether the total charge in your two-object system changed. Use the foil pieces as charge indicators and your metal key to redistribute charge.” Observe for 4 minutes as pairs follow the redistribution step. If a pair stops, ask them to show how they are checking for net charge using the foil pieces.
- 4. In the last 2 minutes ask every pair to hold up both foil pieces and say aloud: “Do we find net charge left on the two-foil system after rubbing and redistribution? Yes or No — why?” Collect one quick sentence from 3–4 pairs to share with the class.

Student instructions:

- 1. Label your two balloons or two objects A and B with the marker. Keep A untouched. Rub B with the wool cloth for 30 seconds. Observe whether A or B attracts the foil pieces, the comb, or the metal key.
- 2. Using the comb: rub the comb with the wool cloth for 15 seconds. Hold the comb near the two foil squares (not touching) and watch whether the foil squares move. Then hold the rubbed key or paperclip near each foil square. Record which object attracts or repels each foil square.
- 3. Redistribute charge test: Tape one aluminum foil square to the table and leave the other free. Touch the rubbed object (B or the comb) to the metal key, then touch that key to the free foil square three times. Next touch the key to the taped foil square three times. Observe whether the two foil pieces show the same behavior (both attracted to the comb or both repelled). Record whether the total visible effect on both foils changed after redistribution.
- 4. On your worksheet slip write: (a) which object gained electrons (felt negative), (b) which lost electrons (felt positive), (c) whether the two-foil system gained or lost net charge after your redistributions, and (d) one sentence: how this supports or opposes the idea that rubbing creates new charge.

Success looks like: Each pair has labeled their two test objects A and B and performed both a rubbing and a comb test (visible rubbed object and untouched object).; Each pair uses the metal key and the two foil squares to demonstrate charge transfer steps (touch sequences) and can point to whether both foil squares show the same sign after redistribution.; At least one member of each pair can state aloud which object gained electrons and which lost electrons, and that the two-foil system did not show a new extra net charge after redistribution.

Support: If you are unsure how to start, follow this worked example sentence: “I rubbed B with wool. B now

attracts small pieces of foil; A does not. That means B gained electrons (more negative) and A lost none. I touched B to the key, then key to free foil three times, then key to taped foil three times. Both foils behaved the same, so total charge did not increase.” Pair a student who is unsure with a student who finished the first rub test.

Extension: Design and describe a quantitative test using only classroom items here to compare the magnitude of charge transferred when rubbing the balloon vs rubbing the comb. Explain what measurable quantity you would use (for example, number of foil leaves separated, angle of deflection estimated visually) and how you would keep units or comparisons explicit (e.g., deflection distance in centimeters or count of attracted foil pieces). Predict the ratio of charges and explain the reasoning from electron transfer at the atomic scale.

Checkpoint Questions

- [understand] You rub a glass rod with silk. At the atomic scale, which particles move and in which direction (qualitatively)?

Expected: Electrons move from one material to the other. Specifically, if glass loses electrons to silk, the glass has fewer electrons than protons and becomes net positive while the silk gains electrons and becomes net negative. (State which material in your classroom example gained electrons.)

- [apply] A metal sphere and a plastic rod are both rubbed and then touched to the floor. Which one is more likely to keep its charge and why, in terms of charge mobility?

Expected: The plastic rod (an insulator) is more likely to keep its charge because charges remain localized; the metal sphere (a conductor) lets charges move and leak to ground through contact so it loses charge. Mention that in the conductor electrons travel to ground but in the insulator they stay stuck where transferred.

- [understand] If you rub two objects together and one becomes $+2\ \mu\text{C}$ and the other $-2\ \mu\text{C}$, what does this tell you about the net charge before and after rubbing? Use conservation of charge in your answer.

Expected: The net charge before and after rubbing is the same. If after rubbing the charges are $+2\ \mu\text{C}$ and $-2\ \mu\text{C}$, their sum is $0\ \mu\text{C}$, so no net charge was created; charge was redistributed (electrons moved) between the objects.

EXIT TICKET

In one short paragraph, explain using atomic-scale evidence why rubbing two materials transfers electrons and does not create net charge. Include which particle moves, how the two objects' signs change, and one concrete classroom example (e.g., comb and paper).

Success indicator: Mentions that electrons (not protons) move from one material to the other; states that one object becomes net negative and the other net positive; explicitly says the algebraic sum of charges is unchanged (gives an example like $+q$ and $-q$ or states net charge remains same); gives a concrete example (comb and paper or glass and silk).

Homework

- Write a one-page explanation (~300 words) using an atomic picture to describe what happens when a nylon comb is rubbed on dry hair. Draw and label a simple diagram showing electrons moving and indicate the final net charge on each object with units.
- Measure and record one real example at home: rub a clean plastic comb on dry hair, bring it near small bits of paper, and note whether paper is attracted. In one paragraph, connect your observation to the atomic-scale electron transfer you drew above.
- Short quiz question (answer on the same sheet): Two identical insulating balloons are

rubbed on wool. Afterward they repel each other. Propose two possible distributions of charge on the balloons (numbers with units) that are consistent with conservation of charge and explain why both are possible.

Estimated time: 40 min

MENTOR MOMENT — THE WORKSHOP COMB SURPRISE

A physics teacher once combed a volunteer's hair in front of the class; the comb picked up bits of paper and stuck to a metal tray until the tray was grounded and the effect vanished.

Takeaway: Seeing a comb attract paper shows charge moved to the comb; touching the metal tray to ground removed the charge—conservation means charge moved away, not disappeared.

Period 2

ENTRY TICKET

You rub a plastic comb on dry hair. Do you think the comb has: (A) gained new charge that did not exist before, (B) lost some charge that disappears, or (C) had electrons transferred from the hair so total charge is conserved? Pick A, B or C and one sentence why.

Expected response: C — electrons are transferred from one object to the other; no new net charge is created.
(Reason: rubbing moves electrons between materials; total charge before and after is the same.)

Teacher Script

0-5 min: Collect entry tickets & announce learning goal

Collect a quick show of hands: how many chose A, B, C? (Pause for hands.) "Good — we will keep that idea in mind. Today we will use the idea that charge is conserved and quantised to find the force between point charges: Coulomb's law. By the end you should be able to explain, with atomic-scale reasoning, why rubbing transfers electrons and does not create net charge."

Board: Write at top: "Learning goal: Use quantisation and conservation to derive Coulomb's law and explain rubbing as transfer of electrons."

- Anticipate: Why must we decide between those three options now?
- Anticipate: What do you mean by 'atomic-scale reasoning'?
- Anticipate: If electrons move, doesn't that create charge somewhere else?

5-13 min: Measured demonstration: identical spheres A,B,C,D

I will show a diagram and talk you through a thought experiment you can imagine as spheres in the lab. Say sphere A has charge q and sphere B has charge q' . Two identical uncharged spheres C and D touch A and B respectively, then are separated so each pair shares charge by symmetry. Ask: "If C touches A and they are identical, what charge does each have afterwards?" (Wait for answers: $q/2$.) "Why? Because charge is conserved and the spheres are identical so charge distributes equally. Same for B and D: q' becomes $q'/2$ on each.)" Then point to the next question: "If the centre-to-centre distance between the original charged spheres A and B is halved, what happens to the force between the charges on A and B if we first replace A and B by the halves $q/2$ and $q'/2$?" Let students estimate before we calculate.

Board: Draw four spheres labelled A(q), C(0) touching then separate $\rightarrow$ A($q/2$), C($q/2$). Similarly B(q'), D(0) $\rightarrow$ B($q'/2$), D($q'/2$). Draw arrow showing original separation r then $r/2$. Write short note: "Charge conserved in contact; identical spheres $\Rightarrow$ equal split."

- Anticipate: Why do the charges split exactly in half? (Because spheres are identical so capacitances equal; by symmetry and conservation charges equalize.)
- Anticipate: Do we need to measure electrons to say $q/2$? (No — conservation + symmetry gives distribution.)
- Anticipate: If they touch briefly, how fast does this happen? (Very quickly compared to our experiment; we treat final distribution only.)

13-23 min: Introduce Coulomb's law (magnitude) by derivation

Write the scalar form we will derive from experiments: $F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$. Now show the calculation for the sphere example. "Original force between A and B: $F = (1/4\pi\epsilon_0) q q' / r^2$."

Then after touching and halving charges and halving separation: compute $F' = (1/4\pi\epsilon_0) (q/2)(q'/2) / (r/2)^2$. Work it out step by step on the board: numerator ($q q'/4$); denominator ($r^2/4$). Simplify: $(q q'/4) / (r^2/4) = q q' / r^2$, so $F' = F$. Say aloud each algebra step and keep units: q in coulombs (C), r in metres (m), force in newtons (N), $1/4\pi\epsilon_0$ has units $\text{N}\cdot\text{m}^2/\text{C}^2$. Conclude: this is consistent with an inverse-square law proportional to product of charges. Say: "This is the same algebra we use to motivate the scalar Coulomb law from experiment."

Board: Write formula stepwise: $F = (1/4\pi\epsilon_0) q q' / r^2$. Then write replacement: $F' = (1/4\pi\epsilon_0) (q/2)(q'/2) / (r/2)^2$.

Show algebraic simplification to $F' = q q' / (4\pi\epsilon_0 r^2)$. Underline units: q (C), r (m), F (N).

- Anticipate: Where did $1/4\pi\epsilon_0$ come from? (It is the experimentally determined constant for vacuum permittivity; we use it as measured. We will not derive its value now.)
- Anticipate: Why does the algebra keep the force the same when both charge halves and distance halves? (Because halving charges multiplies numerator by $1/4$, while halving r squares to $1/4$ in denominator, cancelling.)
- Anticipate: Does this always work for non-spherical objects? (Coulomb's inverse-square form applies to point charges; for extended bodies you must treat charge distribution.)

23-30 min: Derive vector form of Coulomb's law

Introduce the point charge definition: read aloud: "point charge: A charge whose linear size is very small compared to distances considered; all the charge is treated as concentrated at a single point." Draw two point charges q_1 at position r_1 and q_2 at r_2 . Ask: "Which direction is the force on q_1 due to q_2 ?" (Expect answers toward or away along the line joining them depending on signs.) Explain: "The force acts along the line joining the charges; sign of $q_1 q_2$ gives attraction (negative product) or repulsion (positive product)." Write the vector formula: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{12}^2) \hat{r}_{21}$. Define r_{12} as the magnitude of $r_1 - r_2$ and $\hat{r}_{21}$ as the unit vector pointing from 1 to 2? (Clarify: use the source's notation: r_{12} is the distance between charges; $\hat{r}_{21}$ is the unit vector pointing from charge 2 toward charge 1 so direction on force on 1.) Work a quick sign example: $q_1 = +1.0 \times 10^{-6} \text{ C}$ at left, $q_2 = -2.0 \times 10^{-6} \text{ C}$ at right separated by 0.50 m; direction is toward q_2 . Do a short calculation of magnitude with units: compute F magnitude $= (1/4\pi\epsilon_0) (|q_1| |q_2|) / r^2$. Use $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$ if students ask for numbers, but avoid long arithmetic — show units confirm N.

Board: Draw two points labelled r_1 (q_1) and r_2 (q_2). Draw arrow $\hat{r}_{21}$ pointing from 2 to 1. Write definition of point charge. Write vector formula exactly as given in source.

- Anticipate: Why $\hat{r}_{21}$ and not $\hat{r}_{12}$? (Because the formula uses the unit vector pointing from source to the point where force is evaluated; we will stick to the source notation.)
- Anticipate: What if both charges are negative? (Product $q_1 q_2$ positive $\rightarrow$ repulsion; direction given by unit vector still yields correct orientation.)
- Anticipate: Do we need ϵ_0 's numeric value? (Only for numerical answers; for understanding symbolic form we do not compute it now.)

30-36 min: Guided practice: quick problems

Hand out or project two quick prompts. Prompt 1: Two point charges $q_1 = +2.0 \times 10^{-6} \text{ C}$ and $q_2 = +3.0 \times 10^{-6} \text{ C}$ are 0.30 m apart in vacuum. Ask students to estimate whether the force is attraction or repulsion and to set up the magnitude expression. Prompt 2: Two charges $q_1 = +1.0 \times 10^{-6} \text{ C}$ and $q_2 = -1.0 \times 10^{-6} \text{ C}$, 0.20 m apart. Ask students to write the vector form showing direction with unit vector. Walk the room for 3 minutes and listen to setups. Reconvene and write one correct setup for each on the board without doing the full arithmetic (unless a student

volunteers).

Board: Write setups: $F = (1/4\pi\epsilon_0)(2.0\text{e-}6)(3.0\text{e-}6) / (0.30)^2$ N and $F = (1/4\pi\epsilon_0)(1.0\text{e-}6)(1.0\text{e-}6) / (0.20)^2$ N
 $\hat{r}_{21}$ with note: second is attraction so arrow points toward q_2 .

- Anticipate: Should we include signs in the magnitude? (No — magnitude is positive; sign/direction comes from vector form.)
- Anticipate: How do we show the unit vector numerically? (We write the unit vector symbolically for direction; to get components you need coordinates.)
- Anticipate: Can we plug ϵ_0 now? (Yes if you want a numerical answer; not required for the set-up.)

36-40 min: Formative check and consolidate

Ask three quick exit questions out loud and have students write A/B/C on a mini whiteboard: 1) "If you rub two materials, is new net charge created? A Yes B No" (Correct: B, with short explanation: electrons transfer.) 2) "Does Coulomb's magnitude depend on q_1q_2 or q_1+q_2 ? A product B sum" (Correct: A product.) 3) "If distance doubles, how does magnitude change? A doubles B halves C becomes 1/4" (Correct: C becomes 1/4). Collect responses and summarise: "Charge is quantised $q = ne$ and conserved in transfers; Coulomb's law gives magnitude and vector direction along the line between point charges."

Board: Write 3 quick answers with one-line reasons. Final line: "Summary: $q = ne$; $F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$; vector form: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{12}^2) \hat{r}_{21}$."

- Anticipate: What does quantised mean for large objects? (It means total charge is an integer multiple of e ; for macroscopic objects n is very large.)
- Anticipate: Do we ever see exceptions to conservation in rubbing? (No; measurable transfers are transfers of electrons, not creation of charge.)
- Anticipate: Will we use superposition next? (Yes — next period we add contributions from many point charges.)

Blackboard Notes

- Learning goal
- Point charge (definition)
- Quantisation of charge
- Coulomb's law (scalar)
- Coulomb's law (vector)
- Sphere example: A,B,C,D (charge split)
- Quick setups (guided practice)
- Summary
- Learning goal: Use quantisation and conservation to derive Coulomb's law and explain rubbing as transfer of electrons.
- point charge: A charge whose linear size is very small compared to distances considered; all the charge is treated as concentrated at a single point.
- Quantisation of charge: $q = ne$ (q in coulombs, n integer, e = elementary charge)
- Scalar Coulomb's law: $F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$ (F in N, q in C, r in m)
- Vector Coulomb's law: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{12}^2) \hat{r}_{21}$ (direction along line joining charges)
- Sphere example: identical uncharged spheres touching charged spheres -> charges split equally by symmetry ($q \rightarrow q/2$).

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Algebra from example: $F' = (1/4\pi\epsilon_0)(q/2)(q'/2)/(r/2)^2 = (1/4\pi\epsilon_0) q q' / r^2 = F$

- Guided practice setups written (no long arithmetic).
- Summary: $q = n e$; Coulomb magnitude depends on product of charges and inverse square of separation; vector form gives direction and sign (attract/repel).
- Diagram: Four spheres: A(q) touching C(0) → separate → A(q/2), C(q/2). Similarly B(q') and D(0) → B(q'/2), D(q'/2). Show centre-to-centre separation labelled r and r/2.
- Diagram: Two point charges at r₁ (q₁) and r₂ (q₂) with arrow $\hat{r}_{21}$ pointing from charge 2 toward charge 1 to indicate direction of force on 1.
- Diagram: Quick coordinate sketch for guided practice problems showing two charges and separation with labels.
- Formula: $q = n e$
- Formula: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$
- Formula: $\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{21}$

Activities

Counting Electrons and Calculating Coulomb Forces (10 min, problem_set)

Materials: calculator, pencil and eraser, ruler (cm/mm), worksheet (one per student) — printed problems below, periodic table or reference for electron charge value (optional)

1. Hand out the worksheet (student_instructions below) and ask students to work individually. Say: “You have 10 minutes to complete all parts; show units and every algebraic step.” Start the timer.
2. While students work, circulate. For any student stuck at the start, point to the support box on their worksheet and read the first sentence frame aloud once. For students using wrong units, prompt: “What are the SI units for charge, distance, and force?”
3. After 8 minutes, announce: “Two minutes left — check that every step shows units and integer n when you use $q = n \cdot e$.” Continue circulating to note observations listed under success_criteria.
4. At 10 minutes, collect worksheets and make quick notes on common errors (units, non-integer n, sign mistakes) to address in the next whole-class review.

Student instructions:

- Part A — Charge quantisation (4 minutes). A rubber rod is rubbed with fur and ends up with net charge $+3.2 \times 10^{-18} \text{ C}$. Using the electron charge magnitude $e = 1.60 \times 10^{-19} \text{ C}$, answer: (a) Calculate the integer n for the rod, where $q = n \cdot e$. Show every step and include units. (b) Explain in one sentence whether rubbing created new charge or moved electrons between objects, using your n result as evidence.
- Part B — Magnitude form of Coulomb's law (3 minutes). Two point charges are fixed on a lab bench: $q_1 = +4.8 \times 10^{-19} \text{ C}$ (on the rod) and $q_2 = -3.2 \times 10^{-19} \text{ C}$ (on a glass rod). They are separated by 5.0 cm. (a) Using Coulomb's magnitude law $F = k \cdot |q_1 \cdot q_2| / r^2$ with $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$, compute the magnitude of the force in newtons. Show units at each step and convert 5.0 cm to meters before substituting. (b) State whether the force is attraction or repulsion, and justify using the signs of q_1 and q_2 .
- Part C — Vector form direction (3 minutes). Place q_1 at $x = 0 \text{ m}$ and q_2 at $x = +0.05 \text{ m}$ on the x-axis. Using the vector form $F_{12} = k \cdot q_1 \cdot q_2 \cdot (\hat{r}) / r^2$ where $\hat{r}$ points from charge 1 to charge 2, determine the force vector on q_1 due to q_2 . (a) Compute the scalar value (including sign) that multiplies $\hat{r}$. (b) Give the final force as a vector in unit-vector

notation (i.e., $\pm F \hat{i}$) with units.

- **Finish:** Check that all numerical answers include units, that n in Part A is an integer, and that you showed the conversion between cm and m in Part B.

Success looks like: You can observe students writing $q = n \cdot e$ and calculating n as an integer (e.g., $n = +20$) with units C canceling correctly.; You can observe students converting 5.0 cm to 0.050 m before substituting into Coulomb's law and writing the force in newtons (e.g., 1.0×10^{-8} N).; You can observe students indicating attraction or repulsion by citing the opposite or same signs of q_1 and q_2 and writing the force vector with the correct sign on the x-component (e.g., $-F_x \hat{i}$ for attraction toward $+x$).

Support: If you are stuck, start with this sentence frame for Part A: " $q = n \cdot e$, so $n = q/e = (\text{___ C})/(1.60 \times 10^{-19} \text{ C}) = \text{___}$ (no units)." Example worked first item: if $q = +3.2 \times 10^{-18} \text{ C}$ then $n = (+3.2 \times 10^{-18} \text{ C})/(1.60 \times 10^{-19} \text{ C}) = +20$.

Pair with a neighbor who completed Part A for help on unit conversion.

Extension: Suppose instead both charges were positive and fixed at $x = 0$ and $x = +0.05 \text{ m}$. Calculate the net electric field at $x = +0.025 \text{ m}$ (midpoint) due to both charges. Show units and every algebraic step. Then explain qualitatively how the force on a small test charge $+1.0 \times 10^{-9} \text{ C}$ placed at the midpoint would compare to the force on q_1 found earlier.

Checkpoint Questions

- **[understand]** You rub a glass rod with silk and the rod ends up with net $+3e$ (three elementary charges). Explain, in one sentence, what happened to electrons and why no new charge was created.

Expected: Electrons moved from the glass to the silk (or from the silk to the glass depending on materials), leaving the glass deficient by $3e$; charge was transferred between bodies so no new net charge was created — total charge is conserved.

- **[apply]** Two point charges $q_1 = +2e$ and $q_2 = -4e$ are 0.50 m apart in vacuum. Without calculating numbers, state whether the force magnitude on q_1 due to q_2 increases, decreases, or stays the same if q_2 is replaced by two separate point charges each $-2e$ placed at the same location. Explain briefly.

Expected: Stays the same: product $q_1 q_2$ is unchanged because $+2e \times -4e = +2e \times (-2e + -2e)$ summed equals the same total product; Coulomb magnitude depends on the total product and separation, so force magnitude is unchanged.

- **[remember]** Write the vector expression for the force on charge q_1 at position r_1 due to charge q_2 at r_2 , and identify the unit vector direction in words.

Expected: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{12}^2) \hat{r}_{21}$; the unit vector $\hat{r}_{21}$ points from q_1 toward q_2 (i.e., along the line joining them, direction of the force on q_1 due to q_2).

EXIT TICKET

A student says: 'Rubbing a plastic comb on hair creates new charge out of nowhere because the comb becomes charged.' In two sentences, correct the student's statement using atomic-scale charge transfer and state the relevant formula for charge quantisation.

Success indicator: Student states that rubbing transfers electrons between materials (identifying which object loses/gains electrons) so total net charge is conserved, and writes $q = n e$ with n an integer; answer correctly links transfer to conservation rather than creation.

Homework

- 1) Two identical small conducting spheres, A and B, are isolated. A has net charge $+6e$ and B is neutral. The spheres are touched together and then separated. (a) What is the charge on each afterward? Show arithmetic with units. (b) If the separation between A and a fixed third charge C is then halved, explain quantitatively how the Coulomb force between A and C changes (use the scalar Coulomb formula).
- 2) Calculate the magnitude of the electrostatic force between $q_1 = -3e$ and $q_2 = +5e$ separated by 0.20 m in vacuum. Show all steps, carry units (C, m, N), and use $e = 1.602 \times 10^{-19}$ C and $1/4\pi\epsilon_0 = 8.99 \times 10^9$ N·m²/C².
- 3) Draw two point charges of equal magnitude and opposite sign separated by 0.30 m. On the diagram, draw and label the force vector acting on the positive charge due to the negative one (include direction and write the vector form using $\vec{r}$).

Estimated time: 30 min

MENTOR MOMENT — THE SWEATER-AND-BALLOON TRICK

I once rubbed a balloon on a sweater until it stuck to a wall; the balloon lost a tiny number of electrons to the sweater, not 'made' charge. The wall and balloon then attract because of redistributed charges and Coulomb forces.

Takeaway: Friction moves electrons between materials; conservation means rubbing transfers charge, and Coulomb's law predicts the resulting force direction and size.

Period 3

ENTRY TICKET

Two identical neutral spheres C and D are touched to charged spheres A and B respectively. Initially A has charge $+q$ and B has charge $+q'$. After touching, symmetry makes the charges on A+C and B+D become $+q/2$ and $+q'/2$. If the centre-to-centre distance r between A and B is then halved ($r \rightarrow r/2$), how does the magnitude of the electrostatic force between A and B compare to the original force $F = (1/4\pi\epsilon_0) q q' / r^2$? Write one short sentence and a one-line algebraic reason.

Expected response: The force stays the same (unchanged). Algebraic reason: new $F' = (1/4\pi\epsilon_0) (q/2)(q'/2) / (r/2)^2 = (1/4\pi\epsilon_0) q q' / r^2 = F$.

Teacher Script

0-5 min: Activate prior knowledge / Entry ticket review

Collect quick responses (thumbs up for "same", thumbs down for "changes"). Ask one student to read their sentence aloud. If a student gives the "force increases by 4" answer, write that algebra on the board and say: "Hold that — we'll check it". Remind class: we already know Coulomb's law in vector form and that charge is quantised. Say the objective out loud: "Today we will use superposition to add forces, define electric field, and show the field superposes too. By the end you should be able to explain why rubbing transfers electrons rather than creating new net charge."

Board: Write the entry-ticket algebra if a student offers it: $F = (1/4\pi\epsilon_0) q q' / r^2$ and student claim $F' = (1/4\pi\epsilon_0) (q/2)(q'/2) / (r/2)^2 = ?$

- Anticipate: Why do the neutral spheres split charge in half?
- Anticipate: Is the test charge the same as the charged spheres?
- Anticipate: Why did we let them touch—does that change total charge?

5-15 min: Introduce principle of superposition for forces

Say: "We will build the total force from two point charges by adding the vector forces from each. I will show this with a simple two-charge arrangement, then extend to many charges." Draw two fixed source charges Q_1 at $x = -d$ and Q_2 at $x = +d$ on the board (label magnitudes and signs; use $Q_1 = +2 \mu\text{C}$ and $Q_2 = -1 \mu\text{C}$ as a concrete example). Place a small test charge q at $x = 0$ (use $q = +1 \mu\text{C}$). Ask: "What is the force on q from Q_1 ? From Q_2 ?" Have students call out directions; then write each force using the Coulomb-like point-force expression: $F_r = (1/4\pi\epsilon_0)(Q q / r^2) \hat{r}$ (point to directions). Emphasize vectors: show unit vectors from source to test charge, calculate magnitudes with distances (d and d). Then say: "Total force = vector sum of these individual forces" and write the sum. Point out this is the principle of superposition: add each vector contribution.

Board: Draw two charges on horizontal line: label Q_1 at $-d$, Q_2 at $+d$, test charge q at 0 . Write each force term as a vector using the provided Coulomb-like formula and then write total $F = F_1 + F_2$ (vector sum).

- Anticipate: Do we need to use unit vectors every time?
- Anticipate: What if the test charge is negative?
- Anticipate: Does the presence of q change Q_1 and Q_2 ?

15-25 min: Define electric field and its superposition

Say: "Instead of talking about forces on a specific test charge, we define the electric field E that tells us the force per unit test charge at each point. Read the definition with me: 'Test charge: A charge q used to probe the electric field of a source charge; ideally negligibly small so it does not disturb the source charge distribution.'" Write the definition on the board. Then derive the field from the force: place the same Q_1 and Q_2 on the board and say: "Field from a single point source Q at distance r is $E = F/q = (1/4\pi\epsilon_0) Q / r^2$. So the net field is the vector sum of fields from each charge." Emphasize units: show F in newtons (N), q in coulombs (C), so E in N/C. Use the earlier numeric example ($Q_1 = +2 \mu\text{C}$, $Q_2 = -1 \mu\text{C}$, point at $x=0$) and compute E_1 and E_2 numerically (carry units), then add vectors to get net E . Conclude: once we have E , any test charge q experiences $F = q E$. Finally state the superposition principle for E in words: "Net E = sum of E from each source charge."

Board: Write the exact wording of the Test charge definition. Write E from a point charge as derived: $E = (1/4\pi\epsilon_0) Q / r^2$ (derive by dividing F_r by q). Show units: N/C. Then write $E_{\text{net}} = E_1 + E_2$ and plug numeric values for the example, showing arithmetic with units.

- Anticipate: Why divide by q — what if q is not small?
- Anticipate: Is E a physical thing or just a calculation?
- Anticipate: If q is negative, do we still divide by q the same way?

25-33 min: Demonstration and guided practice: field lines and flux

Say: "Now a quick demonstration with field lines and flux to connect the vector field to pictures and to area crossing." Draw a positive point charge and sketch radial arrows outward. Explain: "Electric field lines: curves whose tangent gives field direction; dense lines $\rightarrow$ stronger field; lines start on positive and end on negative charges." Ask a student to come up and draw lines for a positive-negative pair (dipole). Then introduce electric flux wording exactly: read: 'Electric flux: Scalar quantity through an area element ΔS given by the surface dot product $E \cdot \Delta S = E \Delta S \cos\theta$; for a closed surface, area element direction is outward normal.' Show a small flat area element ΔS in the field and draw angle θ between E and normal. Calculate a numeric example: $E = 100 \text{ N/C}$, $\Delta S = 0.02 \text{ m}^2$, $\theta = 60^\circ$, compute $\Delta\Phi = E \Delta S \cos\theta$ with units ($\text{N}\cdot\text{m}^2/\text{C}$).

Board: Draw a positive charge with radial arrows; draw dipole sketch. Draw a little area element with normal, label θ . Write the electric flux element formula exactly as given and do the numeric calculation: $\Delta\Phi = 100 \text{ N/C} \times 0.02 \text{ m}^2 \times \cos 60^\circ = 1.0 \text{ N}\cdot\text{m}^2/\text{C}$.

- Anticipate: Why is flux a scalar if it comes from a vector dot product?
- Anticipate: What does 'density of lines' mean quantitatively?
- Anticipate: What units does flux have—how do they relate to the equation?

33-40 min: Check and consolidate: charge transfer and exit summary

Return to the entry-ticket example. Recompute the algebra on board showing explicitly: $F' = (1/4\pi\epsilon_0) (q/2)(q'/2) / (r/2)^2 = (1/4\pi\epsilon_0) q q' / r^2$. Ask students to explain in one sentence why halving charges and halving distance cancel. Then pose the atomic-scale question to the class: "When two neutral materials are rubbed, why are electrons transferred rather than new charge created?" Prompt with these points to elicit: electrons are mobile, atoms lose or gain electrons, total net charge of system is conserved (electrons move from one object to the other), and protons remain bound in nuclei. Have students write one short answer on paper. Finish by summarising the three core points on the board: superposition for forces, field definition and superposition, and flux element.

Board: Write the algebraic check for the entry-ticket so it is clear. Then list three consolidation bullets: 1) Superposition (forces) — vector sum; 2) Electric field — characteristic of source charges, $E = F/q$ (units N/C); 3)

Flux element $\Delta\Phi = E \Delta S \cos\theta$. Also write the test-charge and electric-flux definitions verbatim for students to copy.

- Anticipate: So charge isn't created when you rub—how do you show that experimentally?
- Anticipate: If electrons move, can the objects end up with more overall net charge than before?
- Anticipate: Do field lines actually exist or are they just a picture?

Blackboard Notes

- Superposition and Electric Field Basics
- Test charge (definition)
- Coulomb-like point-force expression
- Electric field (definition and units)
- Electric field superposition
- Electric field lines and flux
- Test charge: A charge q used to probe the electric field of a source charge; ideally negligibly small so it does not disturb the source charge distribution.
- Superposition (forces): Total electrostatic force = vector sum of individual Coulomb forces from each source charge.
- Electric field E at a point: force per unit test charge; $E = F/q$. Units: N/C.
- Superposition (field): Net E = vector sum of E from each point charge.
- Electric field lines: tangent = field direction; density proportional to strength; start on +, end on -.
- Electric flux through area element ΔS : $\Delta\Phi = E \cdot \Delta S = E \Delta S \cos\theta$. For closed surface, ΔS points outward normal.
- Conservation note (rubbing): Electrons move between materials; protons stay in nuclei; rubbing transfers electrons (net charge redistributed), it does not create net charge.
- Diagram: Two source charges Q_1 at $x=-d$ and Q_2 at $x=+d$ with test charge q at $x=0$; draw force vectors F_1 and F_2 on q .
- Diagram: Single positive point charge with radial field arrows (field lines outward).
- Diagram: Electric dipole sketch (+ and - close together) with field lines and a point in between showing vector addition of fields.
- Diagram: Small flat area element ΔS with normal drawn, angle θ to E vector; label ΔS and θ .
- Formula: $F_r = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \hat{r}$
- Formula: $\Delta\Phi = \mathbf{E} \cdot \Delta\mathbf{S} = E \Delta S \cos\theta$

Activities

Balloon Charge Demo, Field Sketch, and Quick Superposition Calculation (10 min, demonstration)

Materials: 2 balloons (inflated), wool cloth or sweater, 2 lightweight foam or paper balls each tied to a ~30 cm thread (pith-ball substitute) or two small styrofoam balls on strings, marker and whiteboard or chart paper, ruler (30 cm), calculator (class set or teacher-held), printed worksheet with three short prompts (one per pair) — teacher can prepare in 5 minutes

- 1) Grouping and setup (1 minute): Arrange students in pairs. Ask each pair to watch the demonstration and to complete the short worksheet supplied: (A) sketch field lines, (B)

answer a numeric superposition question, (C) compute a simple flux. Hand one worksheet to each pair.

- 2) Demonstration start (2 minutes): At the front, rub both balloons briskly with the wool cloth for 15–20 seconds. Hold the two balloons near each other on strings so students can see whether they attract or repel. Also approach a pith-ball substitute (foam ball on string) to show attraction/repulsion. Say aloud: “We will use this to reason about electron transfer and how fields combine.”
- 3) Explain atomic-scale charge transfer (2 minutes): While students watch the balloons, draw two neutral atoms on the board. Show, step by step, electrons moving from one material to another when rubbed (label electrons with charge $-e$, units coulomb C). Say: “Electrons move; protons stay in the nuclei. Rubbing transfers electrons so one object gains a net negative charge and the other loses electrons and is net positive. The total net charge of the pair equals the difference of electrons moved — electrons are conserved in transfer.”
- 4) Superposition demonstration and quick calculation (3 minutes): Place two point-charge symbols on the board: $q_1 = +3.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $q_2 = -2.0 \mu\text{C}$ at $x = 0.50 \text{ m}$. Tell students you will compute electric field at P at $x = 0.25 \text{ m}$. Write Coulomb's law $E = kq/r^2$ (show units N/C) and compute E_1 from q_1 and E_2 from q_2 with numeric values, carrying units: $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Show vector directions, add them to get net E (show step-by-step numbers and final N/C).
- 5) Field lines and flux explanation (2 minutes): Ask pairs to sketch field lines around a positive and negative charge on their worksheet, emphasizing lines leaving positive and ending on negative and greater density = stronger field. Then demonstrate flux: draw a small area element $\Delta S = 0.020 \text{ m}^2$ at angle $\theta = 30^\circ$ to a uniform $E = 2000 \text{ N/C}$ and compute $\Phi = E \cdot \Delta S = E \Delta S \cos\theta$, showing units ($\text{N}\cdot\text{m}^2/\text{C}$). Ask students to finish their worksheet in the remaining minute.

Student instructions:

- 1) Watch the teacher demonstration closely and note whether the two rubbed balloons attract or repel, and how the foam pith-ball responds.
- 2) On your worksheet Part A: Draw field-line sketches for (i) a single positive point charge and (ii) a positive $+3.0 \mu\text{C}$ and a negative $-2.0 \mu\text{C}$ separated by 0.50 m . Show direction of field vectors at point P located midway ($x = 0.25 \text{ m}$). Label arrows with 'E' and directions.
- 3) Part B (numeric superposition): Use Coulomb's law $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Compute magnitude and direction of E at P from $q_1 = +3.0 \mu\text{C}$ (at $x = 0.00 \text{ m}$) and $q_2 = -2.0 \mu\text{C}$ (at $x = 0.50 \text{ m}$). Show each step with units, then add vectorially to get net E at P in N/C. Write each r in meters and each intermediate result with units.
- 4) Part C (flux): Using $E = 2000 \text{ N/C}$ and area $\Delta S = 0.020 \text{ m}^2$ tilted at $\theta = 30^\circ$, compute the electric flux Φ through ΔS using $\Phi = E \Delta S \cos\theta$. Show units for Φ ($\text{N}\cdot\text{m}^2/\text{C}$).
- 5) Hand your worksheet to the teacher when finished or raise your hand if you need help with any arithmetic step.

Success looks like: Each pair has a field-line sketch showing lines leaving the positive charge and ending on the negative charge, and a direction arrow for E at point P.; On at least four worksheets the teacher sees the step-by-step Coulomb calculations with numbers and units for E_1 and E_2 and a correct vector sum (net E in N/C).; Most pairs have a written value for flux Φ with correct substitution $\Phi = E \Delta S \cos\theta$ and units $\text{N}\cdot\text{m}^2/\text{C}$.

Support: If you are stuck, use this worked first item: For one point charge $q = +1.0 \mu\text{C}$ at the origin, the field at $r = 0.20 \text{ m}$ on the $+x$ axis is $E = kq/r^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \cdot (1.0 \times 10^{-6} \text{ C}) / (0.20 \text{ m})^2 = 224.8 \times 10^3 \text{ N/C} \sim 2.25 \times 10^5 \text{ N/C}$ directed in $+x$. Sentence frame for sketches: 'Field lines leave the positive charge and point _____

(toward/away) from it; at point P the field vector points ____.' If arithmetic is the barrier, pair with a student who can use the calculator while you label directions and units.

Extension: For early finishers: change q_1 to $+5.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and q_2 to $+3.0 \mu\text{C}$ at $x = 0.50 \text{ m}$. Calculate the electric field along the x-axis at $x = 1.00 \text{ m}$ (show all steps and units) and then compute the flux through a square area 0.050 m by 0.050 m placed at $x = 1.00 \text{ m}$ perpendicular to the x-axis if the local field is the field you computed. Explain how the sign of each charge affects the direction of the field contributions.

Checkpoint Questions

- [apply] Two point charges lie on the x-axis: $Q_1 = +4.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $Q_2 = -2.0 \mu\text{C}$ at $x = 0.30 \text{ m}$. A test charge $q = +1.0 \mu\text{C}$ is placed at $x = 0.10 \text{ m}$. Using Coulomb-like point-force expression, find the net electrostatic force vector on q (give magnitude in N and direction along $+x$ or $-x$).

Expected: Compute force from each charge using $F = (1/4\pi\epsilon_0) Q q / r^2$, keep directions. $r_1 = 0.10 \text{ m}$ to the right of Q_1 so F_1 is repulsive $\rightarrow +x$. $r_2 = 0.20 \text{ m}$ to the left of Q_2 so q is to left of Q_2 ; since Q_2 is negative and q positive, force is attractive $\rightarrow +x$ (toward Q_2). Numerical values: $k \sim 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. $F_1 = k(4.0\text{e-}6)(1.0\text{e-}6)/(0.10^2) \sim 3.596 \times 10^{-1} \text{ N}$ to $+x$. $F_2 = k(2.0\text{e-}6)(1.0\text{e-}6)/(0.20^2) \sim 4.495 \times 10^{-2} \text{ N}$ to $+x$. Net $F \sim (0.3596 + 0.04495) \text{ N} \sim 0.4046 \text{ N}$ to $+x$. (Accept answers within few percent.)

- [understand] A small positive test charge is placed at a point in space near several source charges. The class debates whether moving the test charge changes the field produced by the sources. State whether the field produced by the source charges depends on the test charge and give a one-sentence reason using the definition of test charge.

Expected: No — by definition the electric field is a property of the source-charge distribution and a test charge is ideally negligibly small so it does not disturb that distribution; therefore the field is independent of the test charge used to measure it.

- [analyze] At a point exactly halfway between a $+Q$ and a $-Q$ separated by distance d , draw or state the directions of the electric field vectors contributed by each charge and state whether the magnitudes add or cancel. What is the direction of the net field?

Expected: At the midpoint the magnitude of the field from each charge is the same since distances are equal. The $+Q$ produces a field pointing away from $+Q$ (toward the negative charge), and the $-Q$ produces a field pointing toward $-Q$ (also toward the negative charge). The magnitudes add (they point in the same direction), so the net field points from the positive toward the negative charge (along the line joining them).

EXIT TICKET

Explain, using atomic-scale evidence, why rubbing two neutral materials transfers electrons between them and does not create net charge. Keep your answer to three or four sentences.

Success indicator: Mentions that rubbing causes electrons to move from atoms in one material to atoms in the other due to different electron affinity/work function; states that electrons (not protons) move between surfaces; explains that charge is conserved so one object becomes negatively charged and the other positively charged by equal and opposite amounts; links this atomic transfer to macroscopic observation (e.g., one object gains net negative charge and the other equal net positive charge).

Homework

- Problem 1 (calculate, 20 minutes): Three point charges sit on the x-axis: $+6.0 \mu\text{C}$ at $x=0.00 \text{ m}$, $-3.0 \mu\text{C}$ at $x=0.40 \text{ m}$, and $+2.0 \mu\text{C}$ at $x=0.90 \text{ m}$. Find the net electric field vector (magnitude and sign along x) at $x=0.60 \text{ m}$. Show the vector contributions from each charge,

include units, and use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

- Problem 2 (sketch + explain, 10 minutes): Draw the electric field lines for an isolated electric dipole (a small $+Q$ and $-Q$ separated by a short distance). On your sketch, indicate direction of field lines, where line density is largest, and write one sentence explaining what line density means in terms of E-field strength.
- Problem 3 (short written, 15 minutes): In 4–6 sentences, answer the exit-ticket question again: explain why rubbing transfers electrons and not protons, and why this process leaves no net charge created overall. Use the terms 'electron', 'ion', and 'conservation of charge'.

Estimated time: 45 min

MENTOR MOMENT — BALLOON AND HAIR

You rub a balloon on your sweater and hold it near your hair; the hair stands up and the balloon sticks. Electrons moved from the sweater to the balloon, leaving the sweater slightly positive and the balloon negative.

Takeaway: Seeing hair stand up shows electrons moved between materials and that the two objects gained equal and opposite net charge — charge was transferred, not created.

Period 4

ENTRY TICKET

Two point charges, $+Q$ at $x = 0$ and $+Q$ at $x = 4$ cm. A test charge $+q$ is placed at $x = 2$ cm. Draw the direction of the net electric force on $+q$ and say whether the magnitude is zero, nonzero but small, or large. (A sketch and one sentence.)

Expected response: Sketch: arrows from each source pointing away from each $+Q$. The arrows from the two $+Q$'s point in opposite directions at the test charge and have equal magnitude, so they cancel. Net force = zero. Short reason: the distances to each $+Q$ are equal so Coulomb forces are equal and opposite.

Teacher Script

0-5 min: Activate prior knowledge — discuss entry ticket

Say: "Look up here. Five students read your sketch and sentence aloud quickly. Who answered 'zero' for the net force? Who answered 'nonzero'?" Wait for a show of hands. Then say: "Good. One quick reason: for the arrangement $+Q$ at 0 and $+Q$ at 4 cm with $+q$ at 2 cm, why would forces cancel? Who can point to the two forces on the sketch and say which way each arrow points?" If no one volunteers, point yourself: "The force from the left $+Q$ on $+q$ pushes right; from the right $+Q$ it pushes left. Distances are equal so magnitudes are equal. So net force is zero." Then ask: "If the test charge had been $-q$ at 2 cm, what would the net force be?" Wait for responses and correct: "It would also be zero because the magnitudes still match and the directions both point toward the positive sources, producing equal and opposite forces."

Board: Display the same 1D sketch: x-axis, $+Q$ at 0, $+q$ at 2 cm, $+Q$ at 4 cm. Draw two force arrows on $+q$: one right, one left, equal length. Label distances 2 cm and 2 cm. Write one-sentence conclusion: Net force = 0.

- Anticipate: Why do we treat the charges as points instead of small spheres? (Answer: point charge means size $\ll$ distances considered.)
- Anticipate: Does the sign of the test charge always matter? (Answer: it changes direction of each force but not whether magnitudes cancel in this symmetric case.)
- Anticipate: Is electric field the same as force? (Answer: field is force per unit test charge; we'll state unit N/C.)

5-15 min: Demonstration + guided vector Coulomb example

Say: "We'll apply Coulomb's law in vector form to a quick 2D example." Draw on board: point charge $Q = +2.0 \mu\text{C}$ at $r_1 = (0.00 \text{ m}, 0.00 \text{ m})$ and point charge $q = -1.0 \mu\text{C}$ at $r_2 = (0.30 \text{ m}, 0.40 \text{ m})$. Ask: "What is the vector from Q to q , and the separation r ?" Wait for student replies and then compute aloud: $r_{\text{vec}} = r_2 - r_1 = (0.30, 0.40) \text{ m}$. Say: "Compute its magnitude: $r = \sqrt{(0.30)^2 + (0.40)^2} = 0.50 \text{ m}$. Unit vector $\hat{r} = (0.30, 0.40)/0.50 = (0.6, 0.8)$." Next write Coulomb-like point-force expression and apply numbers step by step: $F = (1/4\pi\epsilon_0) Q q / r^2 \hat{r}$. Use $(1/4\pi\epsilon_0) = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Compute scalar factor: $9.0\text{e}9 * (2.0\text{e}-6 * -1.0\text{e}-6) / (0.50^2) =$ compute aloud: numerator $9.0\text{e}9 * -2.0\text{e}-12 = -1.8\text{e}-2 \text{ N}\cdot\text{m}^2 / (0.25 \text{ m}^2) = -7.2\text{e}-2 \text{ N}$. Then multiply by $\hat{r} = (0.6, 0.8)$: $F = (-0.072) * (0.6, 0.8) \text{ N} = (-0.043, -0.058) \text{ N}$. Say: "Interpretation: negative sign means force points opposite our chosen $\hat{r}$ (i.e. toward the source), consistent with an attractive force because q is negative and Q positive." Ask: "Which component is larger in magnitude, x or y ? Do the signs make sense with the sketch?"

Board: Write data: $Q = +2.0 \mu\text{C}$ at $(0,0)$; $q = -1.0 \mu\text{C}$ at $(0.30, 0.40) \text{ m}$. Write $r_{\text{vec}} = (0.30, 0.40) \text{ m}$, $r = 0.50 \text{ m}$,

$\hat{r} \approx (0.6, 0.8)$. Write formula $F = (1/4\pi\epsilon_0) (Q q / r^2) \hat{r}$. Show numeric steps and final vector $F = (-0.043, -0.058) \text{ N}$.

- Anticipate: Why did we pick $\hat{r}$ from Q to q instead of q to Q? (Answer: be consistent; formula as written uses $\hat{r}$ from source to field point; sign of scalar product captures attraction/repulsion.)
- Anticipate: Where did the minus sign come from? (Answer: it comes from q being negative so product Qq is negative.)
- Anticipate: Do units check out? (Answer: yes — show N from the numbers; $1/4\pi\epsilon_0$ has units $\text{N}\cdot\text{m}^2/\text{C}^2$ so after multiplying charges and dividing by m^2 gives N.)
- Anticipate: What if charges are both positive? (Answer: product positive $\rightarrow$ force along $+\hat{r} \rightarrow$ repulsive.)

15-27 min: Teacher-led derivation: dipole far-field (start)

Say: "Now we derive the far-field of a simple dipole and show the $1/r^3$ dependence. We'll use a dipole of $+q$ at $z = +a$ and $-q$ at $z = -a$ on the z -axis. Our field point is at distance r from the origin with $r \gg a$. We'll get the field along the axis first, then the equatorial plane." Draw the dipole: $+q$ at $z=+a$, $-q$ at $z=-a$, field point on $+z$ axis at $z=r$. Ask: "Write the field from each point charge using the point-charge field formula." Write $E_+ = (1/4\pi\epsilon_0) q (r - r_+)/|r - r_+|^3$ and E_- similarly; but for axis we can use magnitudes: distance from $+q$ to field point is $r - a$; from $-q$ is $r + a$. Then expand for $r \gg a$: $1/(r - a)^2 \sim 1/r^2 * (1 + 2a/r)$ and $1/(r + a)^2 \sim 1/r^2 * (1 - 2a/r)$ (explain: use binomial expansion to first order). Substitute and take differences to get net E along axis: $E \sim (1/4\pi\epsilon_0) (2qa / r^3)$ in $+z$ direction. Define dipole moment $p = q(2a)$? Wait — correct definition: $p = q(2a)$ if using $+q$ and $-q$ separated by $2a$ along axis; but more standard is $p = q * (2a)$ in direction from $-$ to $+$. Say: "We express result in terms of p : $E_{\text{axis}} = (1/4\pi\epsilon_0) (2p / r^3)$ in the $+z$ direction." Emphasize units: p has units $\text{C}\cdot\text{m}$, so the combination gives N/C . Then repeat quickly for equatorial point (on plane at distance r from origin, at midplane): contributions are opposite and give $E = -(1/4\pi\epsilon_0) (p / r^3)$ radially inward. Show sign reasoning: along equator, fields from $+q$ and $-q$ have components that cancel along z and add radially inward with net magnitude p/r^3 .

Board: Draw dipole on z -axis: $-q$ at $-a$, $+q$ at $+a$, field point on $+z$ axis at r . Write the point-charge field formula at top. Write expansions: $1/(r \pm a)^2 \sim 1/r^2 (1 \mp 2a/r)$. Show subtraction leading to $E_{\text{axis}} \sim (1/4\pi\epsilon_0)(2qa/r^3)$. Then write $p = q(2a)$ and $E_{\text{axis}} = (1/4\pi\epsilon_0)(2p/r^3)$. Also draw equatorial sketch and final result $E_{\text{equator}} = (1/4\pi\epsilon_0)(-p/r^3)$ (radially).

- Anticipate: Why can we expand $1/(r \pm a)^2$? (Answer: because $r \gg a$, so a/r is small; use binomial expansion to first order.)
- Anticipate: Is $p = q \cdot 2a$ always? (Answer: for this symmetric dipole yes; generally $p = q \cdot d$ where d is vector from negative to positive charge.)
- Anticipate: Why does the field fall like $1/r^3$ instead of $1/r^2$? (Answer: cancellation of leading $1/r^2$ terms from the two opposite charges leaves next-order $1/r^3$ term.)
- Anticipate: What about directions off-axis? (Answer: those require vector addition; we showed axis and equatorial plane which are the canonical results.)

27-35 min: Guided practice: quick calculations and interpretation

Hand out or display two short tasks to do in pairs for 6 minutes. Task A: A dipole with $+q = 2.0 \times 10^{-6} \text{ C}$ at $+0.01 \text{ m}$ and $-q$ at -0.01 m . Find E on axis at $r = 1.0 \text{ m}$ using $E_{\text{axis}} = (1/4\pi\epsilon_0)(2p/r^3)$. Compute p and numeric E and state units. Task B: For same dipole, at equator $r = 1.0 \text{ m}$, compute $E_{\text{equator}} = (1/4\pi\epsilon_0)(-p/r^3)$ and compare magnitude to axis value. Circulate, check arithmetic: $p = q(2a) = 2.0\text{e-}6 * (0.02) = 4.0\text{e-}8 \text{ C}\cdot\text{m}$. Then scalar factor: $1/4\pi\epsilon_0 = 9.0\text{e}9$. For

$r = 1.0 \text{ m}$, $E_{\text{axis}} = 9.0 \times 10^9 \cdot (2 \cdot 4.0 \times 10^{-8}) / (1^3) = 9.0 \times 10^9 \cdot (8.0 \times 10^{-8}) = 0.72 \times 10^2$? Compute carefully with students to reach $E_{\text{axis}} = 720 \text{ N/C}$ along $+z$. E_{equator} magnitude = half that = 360 N/C inward. Ask: "Which direction would a test charge $+1.0 \mu\text{C}$ experience force?" Expect: $F = q_{\text{test}} E$, so $1.0 \times 10^{-6} \text{ C} \cdot E$, numerically small in N.

Board: Write tasks A and B text. Show numeric steps for $p = 4.0 \times 10^{-8} \text{ C}\cdot\text{m}$. Show calculation lines for E_{axis} and E_{equator} and final numeric answers: $E_{\text{axis}} = +720 \text{ N/C}$, $E_{\text{equator}} = -360 \text{ N/C}$ (radial inward). Write $F_{\text{on_test}} = q_{\text{test}} \cdot E$ as reminder.

- Anticipate: Why is equatorial magnitude half the axial? (Answer: derivation gives factor 2 difference; axis is $2p/r^3$, equator is $-p/r^3$.)
- Anticipate: Are these numbers realistic? (Answer: yes; field strengths depend on q and separation; units N/C used.)
- Anticipate: Why multiply by test charge to get force? (Answer: $F = q_{\text{test}} E$; that's the operational definition.)

35-40 min: Formative check + consolidation and link forward

Give two quick oral questions and have students signal answers with thumbs: 1) "If we double r , how does the dipole field at far-field change?" (Expected: decreases by 8, because $1/r^3$.) 2) "Does Gauss's law still hold for a dipole inside a closed surface that encloses no net charge? What is the net flux?" (Expected: flux = 0 because net enclosed charge is zero.) Then say: "Write down the operational definition of electric field and its unit: $E = F/q$, unit N/C. Tomorrow we'll use torque on a dipole in a uniform E and more Gauss law symmetry problems."

Board: Write two quick check questions and answers: 1) $r \rightarrow 2r \Rightarrow E \rightarrow E/8$. 2) Dipole inside closed surface with net charge 0 $\Rightarrow \Phi = 0$. Write operational definition: $E = F/q$, unit N/C. Underline 'useful for torque and Gauss law symmetry next.'

- Anticipate: If flux is zero, does that mean no field inside? (Answer: no — flux zero means net through closed surface is zero; fields can enter and leave.)
- Anticipate: Does $1/r^3$ apply near the dipole? (Answer: no — it's the far-field approximation for $r \gg a$.)
- Anticipate: Will we need the binomial expansion formula? (Answer: yes briefly for derivations; you'll be shown how it's used to first order.)

Blackboard Notes

- Point-charge field and Coulomb force (vector form)
- Dipole configuration and far-field results
- Quick numeric example and definitions
- Point charge (definition): charge concentrated at a point when size $\ll$ distances considered.
- Test charge (definition): small probe charge that does not disturb source distribution.
- Electric field (operational): $E(r) = F(r) / q_{\text{test}}$. Unit: N/C.
- Coulomb-like point-force expression: $F_r = (1/4\pi\epsilon_0) (Q q / r^2) \hat{r}$
- Electric field of a point charge: $E(r) = (1/4\pi\epsilon_0) (Q / r^2) \hat{r}$
- Gauss's law (statement): Electric flux through closed surface = (enclosed charge)/ ϵ_0
- Dipole: $+q$ at $z=+a$, $-q$ at $z=-a$ (separation $2a$). Dipole moment $p = q(2a)$ directed from $-$ to $+$.
- Far-field ($r \gg a$) axial result: $E_{\text{axis}} = (1/4\pi\epsilon_0) (2 p / r^3)$ (along axis from $-$ to $+$).
- Far-field ($r \gg a$) equatorial result: $E_{\text{equator}} = (1/4\pi\epsilon_0) (- p / r^3)$ (radially inward).
- Note: Dipole far-field falls off as $1/r^3$ because leading $1/r^2$ terms cancel between the two charges.

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Force on test charge: $F = q_{\text{test}} E$

- Diagram: 1D sketch: +Q at $x=0$, +q at $x=2$ cm, +Q at $x=4$ cm with equal opposite force arrows on +q.
- Diagram: 2D coordinate sketch: Q at (0,0), q at (0.30,0.40) m with vector $\vec{r}$ drawn and components labeled; show $\hat{r}=(0.6,0.8)$.
- Diagram: Dipole on z-axis: -q at $z=-a$, +q at $z=+a$, field point at $z=r$ on axis; show distances $r-a$ and $r+a$.
- Diagram: Dipole equatorial sketch: field point at distance r in midplane; show vector contributions from +q and -q and resultant.
- Formula: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$
- Formula: $F_r = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \hat{r}$
- Formula: $F(r) = q E(r)$

Activities

Dipole Far-Field Derivation and r^{-3} Check (12 min, simulation)

Materials: whiteboard or chalkboard, calculator (one per student or pair), ruler or straightedge, graph paper or plain paper, pen or pencil, stopwatch or classroom clock, textbook or formula sheet (optional)

1. Group students in pairs. Say: "You will derive the far-field electric field of a simple dipole (+q and -q separated by $2a$) and then check the $1/r^3$ dependence numerically. Work with your partner; one writes and one calculates." Start the stopwatch for 12 minutes and announce time checkpoints at 7 minutes and 2 minutes remaining.
2. On the board, draw the dipole on the x-axis: +q at $x = +a$ and -q at $x = -a$. Write Coulomb's law vector form for a point charge at the origin: $E(r) = (1/4\pi\epsilon_0) Q \hat{r} / r^2$. Tell students they will place the field point on the +x axis at coordinate $x = r$ ($r \gg a$).
3. Tell students to derive the expression for the x-component of the total field on the axis by summing the two contributions using vector form, then expand for $r \gg a$ using the binomial approximation. Circulate while pairs work, watching for correct substitution of position vectors, consistent units (meters, coulombs, N/C), and explicit steps in the binomial expansion. If a pair is stuck, point them to the support scaffold on the worksheet (read it aloud if needed).
4. After 7 minutes, call the class back together for a 2-minute comparison: ask one pair to write the derived far-field expression in terms of dipole moment $p = q(2a)$ or $p = q \cdot (2a)$ on the board and to state the units of p and of E . Confirm the axis result $E_{\text{axis}} = (1/4\pi\epsilon_0)(2p / r^3)$ and identify units (p in C·m, E in N/C).
5. Return students to pairs for the remaining time to compute numerical values: give $q = 1.0 \times 10^{-6}$ C and $a = 0.01$ m. Have them compute E at $r = 0.5$ m and $r = 1.0$ m using both exact Coulomb sums and the far-field approximation; then compare the ratio to check $\sim 1/r^3$ scaling. Collect one completed example from any pair or call on a pair to report their numerical ratio at the end.

Student instructions:

1. Work in pairs. One person writes algebra; the other does arithmetic. We place +q at $x = +a$ and -q at $x = -a$ along the x-axis. Use $q =$ symbol q (keep q general) and $a =$ symbol a (keep a general) for the derivation. Use SI units (q in coulombs, a in metres, r in metres).
2. Write the electric field at a point on the +x axis at position $x = r$ as the vector sum of the fields from each charge. For the +q at $x = +a$, the field at $x = r$ is $E_+ = (1/4\pi\epsilon_0) q (r - a) / (r -$

$a)^2$. For the $-q$ at $x = -a$, the field at $x = r$ is $E_- = (1/4\pi\epsilon_0)(-q) / (r + a)^2$. Show the direction signs explicitly so you have scalar x-components:

- $E_{+,x} = (1/4\pi\epsilon_0) * q / (r - a)^2$ (directed $+x$ if $r > a$)
- $E_{-,x} = (1/4\pi\epsilon_0) * (-q) / (r + a)^2$ (directed $+x$ because negative charge pulls toward $-a$).
- 3. Sum the x-components: $E_x(r) = (1/4\pi\epsilon_0) q [1/(r - a)^2 - 1/(r + a)^2]$. Put over a common denominator and simplify algebraically. Show each algebra step. Replace $q \cdot (2a)$ by p when you reach that product and state p 's units (C·m).
- 4. Assume $r \gg a$. Use the binomial approximation: $1/(r \pm a)^2 \sim 1/r^2 * (1 \mp 2a/r)$. Show the simplification step-by-step and keep units. Derive $E_x \sim (1/4\pi\epsilon_0) (2q \cdot (2a) / r^3) / 2$ if needed and reduce to the canonical form $E_{axis} = (1/4\pi\epsilon_0)(2p / r^3)$. Write $p = q \cdot (2a)$ and verify units: E in N/C.
- 5. Numerical check: using $q = 1.0 \times 10^{-6}$ C and $a = 0.01$ m, compute exact E_x at $r = 0.5$ m by summing the two Coulomb contributions with distances $(r - a)$ and $(r + a)$. Then compute the far-field approximation $E_{axis_approx} = (1/4\pi\epsilon_0)(2p / r^3)$ for the same r . Repeat at $r = 1.0$ m. Compare the ratios Exact/Approx and show whether they approach 1 as r increases. Record your arithmetic and units for each step.

Success looks like: At least one pair has algebra written showing the two scalar x-component terms $1/(r - a)^2$ and $1/(r + a)^2$ and their subtraction.; Pairs carry out the binomial approximation explicitly on paper (showing $1/(r \pm a)^2 \sim 1/r^2 (1 \mp 2a/r)$) and write the resulting $1/r^3$ dependence with $p = q \cdot (2a)$ and units indicated.; Pairs have numeric work visible for $r = 0.5$ m and $r = 1.0$ m showing exact sum and approximate value and a reported ratio Exact/Approx (teacher can read one pair's ratio to confirm values are converging toward 1).

Support: If you are stuck, copy this starting work: $E_x(r) = (1/4\pi\epsilon_0) q [1/(r - a)^2 - 1/(r + a)^2]$. Then form a common denominator: $1/(r - a)^2 - 1/(r + a)^2 = ((r + a)^2 - (r - a)^2) / ((r - a)^2 (r + a)^2)$. Expand the numerator: $(r^2 + 2ar + a^2) - (r^2 - 2ar + a^2) = 4ar$. So $E_x(r) = (1/4\pi\epsilon_0) q (4ar) / ((r - a)^2 (r + a)^2)$. For $r \gg a$, approximate $(r - a)^2 (r + a)^2 \sim r^4$; then $E_x \sim (1/4\pi\epsilon_0) q (4a r) / r^4 = (1/4\pi\epsilon_0) (4qa / r^3)$. Recall $p = q \cdot (2a)$, so this gives $E_x \sim (1/4\pi\epsilon_0)(2p / r^3)$.

Extension: Derive the far-field electric field at an arbitrary point specified by vector r (not on the axis) to show the full angular dependence $E(r) \sim (1/4\pi\epsilon_0)(1/r^3)(3(p \cdot \hat{r}) \hat{r} - p)$. Start from the two-charge vector sum, carry out a series expansion for $r \gg a$ to first order in a/r , and show each vector algebra step, with units carried through. Then compute a numerical example for p along $+x$ and a field point at polar angle $\theta = 60^\circ$ and $r = 1.0$ m using $q = 1.0 \times 10^{-6}$ C and $a = 0.01$ m.

Checkpoint Questions

- [apply] Two point charges, $Q = +3.0 \mu\text{C}$ at position $r_1 = (0.02 \text{ m}, 0, 0)$ and $q = -1.0 \mu\text{C}$ at $r_2 = (0, 0.03 \text{ m}, 0)$. Write the vector expression (magnitude and direction) for the Coulomb force on q due to Q using Coulomb's law in vector form.

Expected: Compute displacement from q to Q : $r = r_1 - r_2 = (0.02, -0.03, 0)$ m. Distance $r = \sqrt{(0.02)^2 + (-0.03)^2} = 0.03606$ m. Unit vector $\hat{r} = (0.02/0.03606, -0.03/0.03606, 0) \sim (0.5547, -0.83205, 0)$.

Coulomb magnitude: $k |Qq| / r^2$ where $k = 1/(4\pi\epsilon_0) \sim 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$. $|F| =$

$8.99 \times 10^9 * (3.0 \times 10^{-6}) * (1.0 \times 10^{-6}) / (0.03606^2) \sim 20.7 \text{ N}$. Sign: Q is positive, q negative $\rightarrow$ attractive, so force on q points toward Q , i.e. $+\hat{r}$. Vector force $F \sim 20.7 \text{ N} * (0.5547, -0.83205, 0) \sim (11.5, -17.2, 0) \text{ N}$.

- [understand] State the operational definition of the electric field at a point and give its SI unit. Then, using that definition, write the relation between force and field for a test charge q placed at that point.

Expected: Electric field E at a point is defined as the force per unit test charge: $E = F/q$ (operationally measured

by placing a small test charge and measuring force). SI unit: newtons per coulomb (N/C). Relation: $F = q E$. (State that the test charge should be negligibly small so it doesn't disturb the source.)

- [analyze] For a dipole consisting of $+q$ at $x = +a$ and $-q$ at $x = -a$, sketch the algebraic steps (no long algebra) that lead to the far-field ($r \gg a$) dependence E proportional to p / r^3 along the axis. Identify where a binomial/Taylor approximation is used and how p appears.

Expected: Start with sum of two point-charge fields on the axis: $E = (1/4\pi\epsilon_0)[q/(r-a)^2 - q/(r+a)^2]$ (along $\hat{x}$). Combine fractions to get numerator $\sim q[(r+a)^2 - (r-a)^2] / (r^2 - a^2)^2 = q(4ar) / (r^4)$ to leading order. Use $r \gg a$ to approximate $(r \pm a)^2 \sim r^2 \pm 2ar$ and denominators $\sim r^4$ (binomial/Taylor expansion neglecting terms $O((a/r)^2)$). Result $E \sim (1/4\pi\epsilon_0)(4aq / r^3) \hat{x} = (1/4\pi\epsilon_0)(2p / r^3) \hat{x}$ since dipole moment $p = q(2a)$ directed along $+x$. This shows E proportional to p / r^3 ; note signs depend on axis side.

EXIT TICKET

A small dipole has charges $+2.0 \mu\text{C}$ at $x = +0.005 \text{ m}$ and $-2.0 \mu\text{C}$ at $x = -0.005 \text{ m}$. Estimate the magnitude and direction of the electric field at a point on the $+x$ axis at $x = 0.20 \text{ m}$ using the far-field dipole formula. Show the key step that yields the $1/r^3$ dependence and express your answer in N/C.

Success indicator: Student computes dipole moment $p = q(2a) = (2.0 \times 10^{-6} \text{ C})(0.01 \text{ m}) = 2.0 \times 10^{-8} \text{ C}\cdot\text{m}$, uses $E_{\text{axis}} \sim (1/4\pi\epsilon_0)(2p / r^3)$ with $r = 0.20 \text{ m}$ and $k = 8.99 \times 10^9$, evaluates $E \sim (8.99 \times 10^9)(2 \times 2.0 \times 10^{-8}) / (0.20^3) \sim$ numerical value $\sim 8.99 \times 10^9 \times 4.0 \times 10^{-8} / 0.008 = (8.99 \times 10^9 \times 5.0 \times 10^{-6}) \sim 45 \text{ N/C}$, and states direction is $+x$ (away from dipole if appropriate). Work shows $1/r^3$ step and units N/C.

Homework

- Problem 1 (20 min): Two point charges $+5.0 \mu\text{C}$ at $(0,0,0)$ and $-3.0 \mu\text{C}$ at $(0.04 \text{ m}, 0, 0)$. Calculate the vector electric field at the point $(0.02 \text{ m}, 0.03 \text{ m}, 0)$. Show displacement vectors, magnitudes, unit vectors, and final vector in N/C.
- Problem 2 (25 min): Derive the far-field electric field (leading term only) on the perpendicular bisector (equatorial plane) of a dipole with charges $+q$ at $+a$ and $-q$ at $-a$. Show the algebra where terms cancel to give $E = -(1/4\pi\epsilon_0)(p / r^3)$ in direction.
- Problem 3 (15 min): Using Gauss's law, explain in one paragraph why the electric field inside a uniformly charged thin spherical shell is zero. Reference the Gaussian surface and enclosed charge explicitly.

Estimated time: 60 min

MENTOR MOMENT — LAB PROBE SURPRISE

In a lab, a student used a too-large test charge and got a different field than theory predicted; after switching to a much smaller probe charge, the measured force matched the calculated $E \cdot q$. The group realized the probe must be negligible.

Takeaway: Always use (or imagine) a test charge small enough that it doesn't change the source distribution; E is defined as force per unit charge.

Period 5

ENTRY TICKET

You have three point charges on a line: $q_1 = +2.0 \mu\text{C}$ at $x = -10 \text{ cm}$, $q_2 = +1.0 \mu\text{C}$ at $x = 0 \text{ cm}$, $q_3 = +2.0 \mu\text{C}$ at $x = +10 \text{ cm}$. A small test charge $q_0 = +0.5 \mu\text{C}$ is placed at $x = 5 \text{ cm}$. In one short sentence, write the direction of the net electrostatic force on q_0 and WHY (no calculation needed).

Expected response: Direction: to the right (positive x). Why: the closer $+2.0 \mu\text{C}$ at $+10 \text{ cm}$ pulls q_0 to the right more strongly than the $+2.0 \mu\text{C}$ at -10 cm pulls to the left, and the $+1.0 \mu\text{C}$ at 0 cm pulls left but weaker.

Teacher Script

0-5 min: Collect entry tickets & quick share

Ask students to hand entry tickets to the edge of their desk. Read one aloud: " $q_1 = +2 \mu\text{C}$ at -10 cm , $q_2 = +1 \mu\text{C}$ at 0 , $q_3 = +2 \mu\text{C}$ at $+10 \text{ cm}$, test $q_0 = +0.5 \mu\text{C}$ at $+5 \text{ cm}$. Which way is the net force on q_0 ?" Then say: "Don't calculate yet — just say direction and why." Call on 2 volunteers who answered differently. If a volunteer says 'left' ask: "Which charge gives the biggest pull on q_0 ? How does distance matter?" If they say 'right' ask: "Which two pulls oppose each other?" Conclude: "Most of you said right because the nearer $+2 \mu\text{C}$ at $+10 \text{ cm}$ dominates. We'll see how to get that answer quantitatively using superposition."

Board: Write the numerical positions and charges from the prompt on the board in a simple line sketch.

- Anticipate: Do we have to include the test charge's own field? (No, we consider forces on q_0 from the other charges.)
- Anticipate: Should we use Coulomb's constant explicitly? (Yes — we'll carry $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ if we calculate.)
- Anticipate: What if the charges were negative? (Direction would reverse for that pull; we'll use sign in vector addition.)

5-15 min: Introduce and state superposition principle

Say: "Superposition: the net force on a charge from several other charges is the vector sum of the forces from each one; other charges don't change each pairwise force." Write the formula-free statement on board. Then do a short worked example step-by-step without pre-calculation: use the entry-ticket configuration. Say: "Let's compute the force on q_0 from each charge and add them as vectors. We'll use Coulomb's law for each pair: $F = k q_0 q / r^2$ and direction along the line between charges." Write k and units on the board. Calculate aloud and on the board: For q_2 at 0 cm : $r = 5 \text{ cm} = 0.05 \text{ m}$, $F_2 = k(0.5\text{e-}6 \text{ C})(1.0\text{e-}6 \text{ C})/(0.05^2)$. For q_3 at $+10 \text{ cm}$: $r = 0.05 \text{ m}$, $F_3 = k(0.5\text{e-}6)(2.0\text{e-}6)/(0.05^2)$. For q_1 at -10 cm : $r = 0.15 \text{ m}$, $F_1 = k(0.5\text{e-}6)(2.0\text{e-}6)/(0.15^2)$. Evaluate each magnitude using $k = 8.99 \times 10^9$ and show arithmetic steps, carrying units N . Then assign signs: right = $+x$, left = $-x$. Add $F = F_3 + F_2(-) + F_1(-)$ with their signs to get net. Point out which term dominated. Ask students to check arithmetic in pairs for 2 minutes.

Board: Write stepwise Coulomb magnitudes with numbers substituted and carry units; at the end write the net force numerically with units and a $+x$ or $-x$ sign.

- Anticipate: Do we need vector components here? (Yes if not collinear; here all are on x so we use sign convention.)
- Anticipate: Why can we add forces linearly? (Because superposition says contributions are

independent.)

- Anticipate: Do we include forces between the other charges? (Not for the net force on q_0 ; only forces on q_0 from each other charge.)

15-23 min: Demonstration: dipole torque derivation

Say: "Now we derive the torque on a dipole in a uniform field. I'll use a simple permanent dipole: $+q$ and $-q$ separated by distance d , with the dipole centred at origin and the vector from $-q$ to $+q$ called s . The dipole moment is $p = q s$ (direction from negative to positive)." Draw the two charges and label forces from a uniform E field pointing up. Then calculate the forces: top charge feels $F = qE$ upward (if $+q$) or downward (if $-q$) — be explicit: the $+q$ feels $F_+ = q E$, the $-q$ feels $F_- = -q E$. Show that net force $F_+ + F_- = 0$. Next compute torque about the centre: write $\tau = r_+ \times F_+ + r_- \times F_-$. Show algebra that simplifies to $\tau = p \times E$. Keep units: p in $C \cdot m$, E in N/C so τ in $N \cdot m$. Point at the written final formula (use the source formula exactly). Then ask: "If E is uniform there is no net force — what if E varies across the dipole?" (briefly note there can be net force — we do not derive that now.)

Board: Draw dipole cartoon: two charges separated by vector s , show r_+ and r_- from centre, write F_+ and F_- , show cross-product algebra leading to $\tau = p \times E$ and label units.

- Anticipate: Why is p defined from negative to positive? (By convention, so sign of p encodes orientation.)
- Anticipate: How can forces cancel but a torque remain? (Because forces act at different positions producing a net moment.)
- Anticipate: If the field is non-uniform, will there be a net force? (Yes — briefly acknowledge and promise a later example.)

23-29 min: Introduce continuous charge densities (λ , σ , ρ) and electric flux element

Say: "When charge is spread out, we define densities: linear λ , surface σ , volume ρ as charge per length, area, volume. We'll use the exact wording from the text." Read and write each definition exactly: $\sigma = \Delta Q / \Delta S$, $\lambda = \Delta Q / \Delta l$, $\rho = \Delta Q / \Delta V$. For each, state SI units aloud and write them: λ in $C \cdot m^{-1}$, σ in $C \cdot m^{-2}$, ρ in $C \cdot m^{-3}$. Then introduce electric flux for a small area: $\Delta \Phi = E \cdot \Delta S$ and say the flux through a closed surface sums to q_{enc} / ϵ_0 — write Gauss's law (integral form) on the board and read it aloud, naming ϵ_0 units $C^2 / (N \cdot m^2)$ if asked. Give a quick physical anchor: "If a small area faces the field, $E \cdot \Delta S$ equals how many field lines pierce that area weighted by strength."

Board: Write the three density definitions with units and write $\Delta \Phi = E \cdot \Delta S$ and the Gauss's law formula exactly as given in the source.

- Anticipate: Why are the deltas used instead of limits? (We use macroscopic averages; later we'll take limits to define continuous densities.)
- Anticipate: Are units always C/m^n ? (Yes: length power matches dimension.)
- Anticipate: Does ΔS have direction? (Yes: ΔS is a vector normal to the area; that's why we use the dot product.)

29-37 min: Guided practice: identify densities and compute ΔQ

Project or draw three small quick prompts and read them one at a time. Prompt A: "A thin wire has total charge $Q = 6.0 \times 10^{-6} C$ uniformly distributed along length $L = 2.0 m$. What is λ and its unit?" Prompt B: "A square metal plate $0.10 m$ by $0.10 m$ has total charge $Q = 1.0 \times 10^{-8} C$ uniformly on its surface. What is σ ?" Prompt C: "A small cube of insulating material of side $2.0 cm$ contains total charge $Q = 4.0 \times 10^{-9} C$ uniformly. What is ρ ?" For each, give students 90

seconds to compute in pairs, then call a pair to report and write results on the board. Insist on units written explicitly. Walk around and check one or two groups, correcting arithmetic if needed.

Board: Write each prompt and then write the computed λ , σ , ρ with units after students report.

- Anticipate: Do we round constants? (Use the given numbers; keep units consistent.)
- Anticipate: For the cube, do we use m^3 for volume? (Yes — convert cm to m before computing.)
- Anticipate: What if distribution isn't uniform? (Then λ , σ , ρ vary with position; these formulas give local/average values.)

37-40 min: Quick check & consolidation

Ask three rapid oral questions and have students thumb-signal (1 = A, 2 = B, 3 = C) to show answers. Q1: "If two equal positive charges are equidistant from a test charge, will their pulls cancel?" Q2: "Unit of surface charge density σ ?" Q3: "Does a dipole in a uniform field experience net force or net torque or both?" After each thumbs, call on a student to justify briefly (one sentence). Close: "Next time we'll use these ideas to compute fields of an infinite wire and plane; today we made superposition and densities concrete."

Board: Underline the key final results on the board: superposition statement, definitions, Gauss's law, $\tau = p \times E$.

- Anticipate: Will we need long algebra for wire/plane examples? (We'll keep symmetry to make integrals simple.)
- Anticipate: Can you show the dipole torque again? (Offer to re-draw after class if needed.)

Blackboard Notes

- Superposition (verbal)
- Worked example: forces on q_0
- Dipole torque
- Charge densities (definitions and units)
- Electric flux and Gauss's law
- Superposition: Net force on a charge = vector sum of forces from each other charge (each pairwise force is unchanged by others).
- Coulomb's law used per pair: $F = k q_{\text{test}} q_i / r^2$ (use vector direction; $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$).
- Entry-ticket line: $q_1 = +2.0 \mu\text{C}$ @ -0.10 m , $q_2 = +1.0 \mu\text{C}$ @ 0 m , $q_3 = +2.0 \mu\text{C}$ @ $+0.10 \text{ m}$, $q_0 = +0.5 \mu\text{C}$ @ $+0.05 \text{ m}$.
- Compute magnitudes: $F_i = k(0.5 \times 10^{-6} \text{ C})(q_i)/(r_i^2)$; assign sign: right = $+x$, left = $-x$; net $F = \sum F_i$ (with signs) $\rightarrow$ result in N.
- Dipole: $p = q s$ (from negative to positive). Forces on $+q$ and $-q$ in uniform E cancel: $F_{\text{net}} = 0$, but produce torque.
- Torque on electric dipole: $\tau = p \times E$ (units: $\text{C}\cdot\text{m} \times \text{N/C} = \text{N}\cdot\text{m}$).
- Definitions (exact wording): $\sigma = \Delta Q/\Delta S$ (surface charge density), unit: $\text{C}\cdot\text{m}^{-2}$.
- $\lambda = \Delta Q/\Delta l$ (linear charge density), unit: $\text{C}\cdot\text{m}^{-1}$.
- $\rho = \Delta Q/\Delta V$ (volume charge density), unit: $\text{C}\cdot\text{m}^{-3}$.
- Small flux element: $\Delta\Phi = E \cdot \Delta S$ (ΔS is area vector).
- Gauss's law (integral, closed surface): closed integral $E \cdot dS = q_{\text{enc}} / \epsilon_0$
- Diagram: Number line showing q_1 at -0.10 m , q_2 at 0 , q_3 at $+0.10 \text{ m}$, q_0 at $+0.05 \text{ m}$ with arrows showing individual forces on q_0 .

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Diagram: Dipole cartoon: $-q$ and $+q$ separated by vector s , centre marked, r_+ and r_- vectors, uniform E shown as parallel upward arrows, show F_+ and F_- and torque sense (curved arrow).

- Diagram: Small area element ΔS with normal vector and field vector E showing dot product for $\Delta\Phi = E \cdot \Delta S$.
- Formula: $\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{q_{\text{enc}}}{\epsilon_0}$
- Formula: $\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$

Activities

Three Charges, a Patch, and a Dipole: Quick Think-Pair-Share (10 min, think_pair_share)

Materials: whiteboard or chart paper, marker, calculator (optional), printed one-page prompt per student (teacher can copy the student_instructions text)

1. Hand each student a printed prompt (or project the student_instructions). Tell students: "You have 3 minutes to work silently and answer the three numbered items for yourself." Start a visible 3-minute timer.
2. After 3 minutes, ask students to pair up with a neighbor. Say: "You have 4 minutes to compare answers, agree on one written solution per pair, and write any corrected steps on your paper or the whiteboard." Circulate and listen for reasoning; when you hear an incorrect step, point them to re-check vector directions or units but do not give the answer.
3. After 4 minutes of pair work, call for two pairs to share their solutions on the board (1 pair for the force question, 1 pair for densities/flux/torque). Allow 3 minutes total for the two groups to present and for you to make one clear correction or highlight one correct derivation step.

Student instructions:

- Think phase (3 min, individual): Answer the three numbered tasks below. Show units and at least one short derivation step for each numeric result.
- 1) Superposition force (do the vector addition step): Three point charges lie on the x-axis: $q_1 = +2.0 \mu\text{C}$ at $x = 0 \text{ m}$, $q_2 = -1.0 \mu\text{C}$ at $x = 0.30 \text{ m}$, and $q_3 = +1.0 \mu\text{C}$ at $x = 0.50 \text{ m}$. Find the net electrostatic force on q_3 . Use Coulomb's law $F = k q_i q_j / r^2$, with $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Show directions as $+x$ or $-x$ and carry units (N). Write the two forces on q_3 from q_1 and q_2 , then add them.
- 2) Small-area flux (one short dot product): At the location of q_3 , imagine a small flat square patch of area $A = 1.0 \times 10^{-4} \text{ m}^2$ whose area vector points in the $+x$ direction. Using the net electric field at q_3 found by dividing your net force by q_3 ($E_{\text{net}} = F_{\text{net}} / q_3$), compute the electric flux through the patch: $\Phi = E_{\text{net}} \cdot A = E_{\text{net}} \cdot A \cdot \cos\theta$. State θ and units ($\text{V}\cdot\text{m}$ or $\text{N}\cdot\text{m}^2/\text{C}$).
- 3) Densities and dipole torque (definitions and a quick torque calc): (a) Give the definition and SI unit for linear charge density λ , surface density σ , and volume density ρ , each in one short sentence (include symbol, formula relating a small element of charge dq , and unit). Example format: $\lambda = dq/dl$, unit $\text{C}\cdot\text{m}^{-1}$. (b) A dipole with moment $p = 4.0 \times 10^{-29} \text{ C}\cdot\text{m}$ is in a uniform electric field $E = 2.0 \times 10^5 \text{ N/C}$ directed $+y$. Calculate the magnitude of the torque τ when the dipole is oriented at 30° from $+y$. Show the vector or magnitude step from $\tau = p \times E$ and give units ($\text{N}\cdot\text{m}$).
- Pair phase (4 min): Compare your answers; if you disagreed, re-calculate the numeric steps together. Agree on one clear line-by-line solution to write down.

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Share phase (3 min): Be ready to present your pair's solution for the part assigned by the teacher.

Success looks like: Circulating teacher can see each pair write the two individual Coulomb forces on q_3 with signs (+x or -x) and numeric values in newtons before summing them.; Each pair computes E_{net} at q_3 with units N/C and uses it to calculate flux Φ through the small patch, showing the angle θ used and the product $E_{\text{net}} \cdot A$ with units.; On the densities/torque item, each pair writes the three density definitions with correct units ($\text{C}\cdot\text{m}^{-1}$, $\text{C}\cdot\text{m}^{-2}$, $\text{C}\cdot\text{m}^{-3}$) and shows the torque magnitude computed from $\tau = p E \sin\theta$ with units N·m.

Support: If you are stuck: (a) Start by computing the two distances to q_3 : $r_{13} = 0.50$ m and $r_{23} = 0.20$ m. (b) Compute one Coulomb force first: $F_{13} = k \cdot q_1 \cdot q_3 / r_{13}^2$; sign: q_1 and q_3 are both positive so force on q_3 from q_1 is +x (repulsive). Work that number out first, then compute F_{23} and remember q_2 is negative so it attracts q_3 (direction -x). For the torque item, use the numerical form $\tau = p E \sin(30^\circ) = p E \cdot 0.5$ as the worked first step.

Extension: If finished early: (a) Recompute the net force on q_3 if q_2 is moved to $x = 0.40$ m. Show how the signs and magnitudes change and explain physically why. (b) For the flux item, compute the flux through a patch whose area vector makes 60° with E_{net} and compare magnitudes. (c) For the dipole, write the full vector torque τ (use unit vector notation) when p points at 30° above +y in the xy-plane and $E = 2.0 \times 10^5$ N/C in +y.

Checkpoint Questions

- [apply] Two point charges lie on the x-axis: $q_1 = +3.0 \mu\text{C}$ at $x = 0.20$ m and $q_2 = -2.0 \mu\text{C}$ at $x = 0.50$ m. A test charge $q_3 = +1.0 \mu\text{C}$ is at $x = 0.35$ m. Using superposition, find the net electrostatic force on q_3 (give magnitude, sign, and units). Show the vector (one-dimensional) addition and the electric constant you use.

Expected: Compute force from q_1 on q_3 : $F_{13} = k q_1 q_3 / r_{13}^2$ to the right (positive x) because both positive. $r_{13} = 0.35 - 0.20 = 0.15$ m. $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. $F_{13} = (8.99 \times 10^9)(3.0 \times 10^{-6})(1.0 \times 10^{-6}) / (0.15^2) \sim 1.20$ N to the +x. Force from q_2 on q_3 : $F_{23} = k q_2 q_3 / r_{23}^2$; q_2 is negative so attract — direction to the +x if q_2 is to right of q_3 ? Here q_2 at 0.50 m, q_3 at 0.35 m, so attraction is to the +x. $r_{23} = 0.50 - 0.35 = 0.15$ m. Magnitude $F_{23} = (8.99 \times 10^9)(2.0 \times 10^{-6})(1.0 \times 10^{-6}) / (0.15^2) \sim 0.80$ N to the +x. Net $F = F_{13} + F_{23} = 1.20 \text{ N} + 0.80 \text{ N} = 2.00 \text{ N}$ in +x direction. Teacher can mark correct units (N), correct use of k , and correct signs/directions.

- [remember] Write the macroscopic definitions and SI units for linear, surface, and volume charge densities (λ , σ , ρ). For each, give a concrete physical example (one sentence) with numbers and units.

Expected: λ (linear charge density) = $\Delta Q / \Delta l$, unit $\text{C}\cdot\text{m}^{-1}$. Example: a 2.0 m wire carrying total charge $+6.0 \mu\text{C}$ has average $\lambda = 6.0 \times 10^{-6} \text{ C} / 2.0 \text{ m} = 3.0 \times 10^{-6} \text{ C}\cdot\text{m}^{-1}$. σ (surface charge density) = $\Delta Q / \Delta S$, unit $\text{C}\cdot\text{m}^{-2}$. Example: a square metal plate $0.10 \text{ m} \times 0.10 \text{ m}$ carrying $-1.0 \mu\text{C}$ has $\sigma = -1.0 \times 10^{-6} \text{ C} / 0.01 \text{ m}^2 = -1.0 \times 10^{-4} \text{ C}\cdot\text{m}^{-2}$. ρ (volume charge density) = $\Delta Q / \Delta V$, unit $\text{C}\cdot\text{m}^{-3}$. Example: a small sphere of volume $4.0 \times 10^{-6} \text{ m}^3$ with total charge $+2.0 \mu\text{C}$ has $\rho = 2.0 \times 10^{-6} \text{ C} / 4.0 \times 10^{-6} \text{ m}^3 = 0.50 \text{ C}\cdot\text{m}^{-3}$. Teacher checks for correct formulas, units, and plausible numeric examples.

- [apply] A permanent electric dipole with dipole moment $p = 4.0 \times 10^{-29} \text{ C}\cdot\text{m}$ is placed in a uniform electric field $E = 2.0 \times 10^5 \text{ N}\cdot\text{C}^{-1}$. The angle between p and E is 60° . (a) What is the magnitude of the torque on the dipole? (b) Does the dipole experience a net force? Explain in one sentence.

Expected: (a) Torque magnitude $\tau = p E \sin\theta$. $\tau = (4.0 \times 10^{-29} \text{ C}\cdot\text{m})(2.0 \times 10^5 \text{ N}\cdot\text{C}^{-1})\sin 60^\circ = (8.0 \times 10^{-24} \text{ N}\cdot\text{m})(\sqrt{3}/2) \sim 6.9 \times 10^{-24} \text{ N}\cdot\text{m}$. Units N·m. (b) In a uniform field the net force is zero because equal and opposite forces on the + and - charges sum to zero; only a torque acts. Teacher checks numerical value with units and the short explanation that uniform E gives no net force.

EXIT TICKET

A charge $q_A = +4.0 \mu\text{C}$ sits at $(0, 0) \text{ m}$ and $q_B = -1.0 \mu\text{C}$ sits at $(0.30, 0) \text{ m}$. A third charge $q_C = +2.0 \mu\text{C}$ is at $(0.30, 0.40) \text{ m}$. (1) Using vector superposition and $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, calculate the net electrostatic force vector on q_C (give components in N). Show the two force vectors you add. (2) Write the formula and SI unit for surface charge density σ .

Success indicator: Correct computation of the two force vectors (from q_A and q_B) with direction and components, correct use of distances (include $\sqrt{0.30^2 + 0.40^2}$ where needed), numeric addition giving net F_x and F_y in newtons with k shown; and correct definition $\sigma = \Delta Q / \Delta S$ with units $\text{C}\cdot\text{m}^{-2}$.

Homework

- Compute the net force on a $+1.0 \mu\text{C}$ charge located at the origin from three charges at $(0.10 \text{ m}, 0)$, $(-0.10 \text{ m}, 0)$, and $(0, 0.15 \text{ m})$. Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Show vector components and final magnitude and direction (degrees).
- A thin insulating rod 0.50 m long carries total charge $+9.0 \mu\text{C}$ uniformly distributed. (a) Compute the linear charge density λ and give its unit. (b) If you take a 0.10 m piece of the rod, what charge does that piece carry? Show units.
- Sketch a small diagram and explain in two sentences why a permanent dipole in a strictly uniform electric field feels a torque but no net force. Reference the forces on the positive and negative charges and show how they add.

Estimated time: 30 min

MENTOR MOMENT — LAB DAY SURPRISE: VECTOR ADDITION MATTERS

In a lab, my partner and I measured forces between three charged beads. We first added magnitudes and got nonsense; only when we added vectors component-wise did the balance of forces make sense and the beads stayed where predicted.

Takeaway: Always resolve electrostatic forces into components and add those components — correct directions matter as much as magnitudes.

Period 6

ENTRY TICKET

You have a long straight wire with uniform linear charge density $\lambda = +2.0 \times 10^{-6} \text{ C/m}$. At a point 0.10 m from the wire you measure the electric field to point radially outward. Which of these two expressions could be the magnitude of that field? Circle one and quickly write one sentence explaining your choice.

A) $E = \lambda/(2\epsilon_0)$

B) $E = \lambda/(2\pi \epsilon_0 r)$

(Answer in 3 minutes.)

Expected response: Students should circle B and write: "B has r in it so the field decreases with distance; A has no r so it can't depend on distance. For $r=0.10 \text{ m}$, B gives a number with units N/C ."

Teacher Script

0-5 min: Collect entry tickets; set goal

Turn to me when the 3 minutes are up. Collect one sentence answers from three different students: one who chose A, one who chose B, and one explanation that mentioned distance. Say: "Today we will use Gauss's law to derive the formulas you just guessed at. By the end you must be able to show the steps and explain the units. Keep your calculators handy; carry units through each calculation."

Board: Write the day's objectives and the three target charge geometries: infinite wire (λ), infinite plane (σ), thin spherical shell (q).

- Anticipate: Why would anyone pick A? (students noticing A has no r)
- Anticipate: Do we need to plug numbers in now?
- Anticipate: What does ϵ_0 mean again?

5-15 min: Derive field of infinite straight wire

Say: "We start with an infinitely long straight wire with uniform linear charge density $\lambda \text{ (C/m)}$. Choose a cylindrical Gaussian surface coaxial with the wire of radius r and length L . Ask: 'What symmetry tells us about the direction of E and its dependence on position around the cylinder?' Wait for 'radial, same at every point on the curved surface.' Then write Gauss's law on the board: closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0$. Point to the curved side: 'Only the curved side contributes because on the end caps E is parallel to the axis and flux is zero if wire is along the axis.' Compute flux: $E(r) \times (\text{curved surface area}) = E(r) \times (2\pi r L)$. Compute enclosed charge: $Q_{\text{enc}} = \lambda L$. Solve for $E(r)$: $E(r) = \lambda/(2\pi \epsilon_0 r)$. Emphasize units: λ in C/m , r in m , ϵ_0 in $\text{C}^2/(\text{N}\cdot\text{m}^2)$ so E in N/C . Draw example: $\lambda = 2.0 \times 10^{-6} \text{ C/m}$ at $r=0.10 \text{ m}$ and compute E numerically: $E = (2.0 \times 10^{-6})/(2\pi \times 8.85 \times 10^{-12} \times 0.10) \text{ N/C}$; work the arithmetic on the board and give the value. Ask: 'Which direction is the vector?' and answer: radially outward (unit vector $\hat{n}$).

Board: Write: Assume cylinder radius r , length L . Then Flux = $E(r) \cdot (2\pi r L)$. $Q_{\text{enc}} = \lambda L$. Gauss: $E(2\pi r L) = \lambda L/\epsilon_0 \rightarrow E(r) = \lambda/(2\pi \epsilon_0 r) \hat{n}$. Show numeric example substitution and final N/C value.

- Anticipate: Why ignore the end caps? (students about field direction)
- Anticipate: What if the wire isn't infinite? (edge effects)

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Anticipate: Do we have to pick L? (it cancels)

15-23 min: Derive field of infinite plane sheet

Say: "Now an infinite planar sheet with uniform surface charge density σ (C/m²). Use a small cylindrical pillbox Gaussian surface that intersects the plane: one flat face above the plane, one below. Ask: 'What is the field direction and how does symmetry help?' Guide to answer: field is perpendicular to the plane and same magnitude on both sides. Write Gauss's law: closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0$. The flux is $E \cdot A$ (top) + $E \cdot A$ (bottom) = $2EA$ because field is normal and same magnitude on both sides. Enclosed charge is σA (charge on the patch of plane inside the pillbox). So $2EA = \sigma A/\epsilon_0 \rightarrow E = \sigma/(2\epsilon_0)$. Emphasize units: σ in C/m² gives E in N/C. Ask: 'What changes if σ is negative?' Students should say 'direction reverses but magnitude same.'

Board: Draw the plane and pillbox. Write: Flux = $2EA$. $Q_{\text{enc}} = \sigma A$. $2EA = \sigma A/\epsilon_0 \rightarrow E = \sigma/(2\epsilon_0) \hat{n}$ (direction $\pm$ depending on sign of σ).

- Anticipate: Why does the field not depend on distance? (students surprised)
- Anticipate: What about near the edge of a finite sheet? (edge effects)
- Anticipate: Does thickness matter? (if thin sheet vs slab)

23-31 min: Derive field of a thin spherical shell (inside and outside)

Say: "Finally, a thin spherical shell of radius R carrying total charge q uniformly. Use a spherical Gaussian surface of radius r centered on the shell. Ask: 'What is E 's direction and dependence on angle?' Students: radial and same at each point on a sphere. Write Gauss: closed integral $E \cdot dA = E(r) \cdot 4\pi r^2$. Case 1: $r > R$. Then $Q_{\text{enc}} = q$. So $E \cdot 4\pi r^2 = q/\epsilon_0 \rightarrow E = q/(4\pi \epsilon_0 r^2) \hat{r}$. Case 2: $r < R$. The Gaussian surface encloses no charge, $Q_{\text{enc}} = 0$, so $E = 0$. Emphasize units: q in C, r in m, gives N/C. Point to the earlier example in the source: 'outside looks like a point charge at centre.'

Board: Draw sphere and two Gaussian radii $r > R$ and $r < R$. Write: For $r > R$: $E(4\pi r^2) = q/\epsilon_0 \rightarrow E = q/(4\pi \epsilon_0 r^2) \hat{r}$. For $r < R$: $Q_{\text{enc}} = 0 \rightarrow E = 0$.

- Anticipate: Why is there no enclosed charge for $r < R$ if shell has charge? (students clarifying 'thin shell' assumption)
- Anticipate: Does this work if charge is not uniform? (only by symmetry)
- Anticipate: What about inside a solid charged sphere? (not covered here)

31-36 min: Paired practice problems

Say: "Turn to a partner. I'll hand out two quick problems: (1) Compute E at $r = 0.05$ m from a wire with $\lambda = 5.0 \times 10^{-7}$ C/m. (2) Compute E just outside a spherical shell of radius 0.20 m with $q = 4.0 \times 10^{-6}$ C. Show units in each step. You have 4 minutes; I'll circulate." After 3 minutes, call time and ask one pair to present their steps for problem (1).

Board: Write the two practice problems with given numbers and remind: show units.

- Anticipate: Should we use the formulas we just derived? (yes)
- Anticipate: Can we approximate π as 3.14? (yes, but mention calculator okay)
- Anticipate: Do we include direction? (yes: radial outward if charge positive)

36-40 min: Check and consolidate; final summary

Ask: "In one sentence each, tell me the dependence of E on r for the wire, plane, and shell (inside and outside)." Call on three students. Then read the blackboard formulas aloud and say: "These are the results you must be able to derive with Gauss's law. Remember: symmetry chooses your Gaussian surface; units must check out; and edge effects break the ideal results for finite objects." Assign: 'If you finish early, re-derive one of the results at home and check units.'

Board: Underline the three final formulas and add a short checklist: symmetry, Gaussian surface, compute flux, compute Q_{enc} , solve for E , check units.

- Anticipate: Can we memorize the formulas? (yes, but know the derivations)
- Anticipate: What exactly is ϵ_0 ? (permittivity of free space; value if they ask)
- Anticipate: Will these be on the test? (yes, derivations expected)

Blackboard Notes

- Today's objectives
- Infinite line charge (λ)
- Infinite plane sheet (σ)
- Thin spherical shell (q)
- Checklist for Gauss's-law derivations
- Use Gauss's law: closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0$
- Choose Gaussian surface by symmetry; carry units through every step
- Infinite wire: cylindrical surface, radius r , length $L \rightarrow Q_{\text{enc}} = \lambda L$
- Infinite plane: pillbox area $A \rightarrow Q_{\text{enc}} = \sigma A$
- Spherical shell: spherical surface radius $r \rightarrow Q_{\text{enc}} = q$ for $r > R$, $Q_{\text{enc}} = 0$ for $r < R$
- Diagram: Cylindrical Gaussian surface around a straight wire: label radius r and length L ; show E arrows radially outward on curved surface and perpendicular to end caps
- Diagram: Pillbox straddling an infinite plane: draw top and bottom faces area A , show E arrows perpendicular on both sides
- Diagram: Spherical shell with two Gaussian surfaces: one $r > R$, one $r < R$; show radial E arrows outside and none inside
- Formula: $E(r) = \frac{\lambda}{2\pi\epsilon_0 r}, \hat{n}$
- Formula: $E = \frac{\sigma}{\epsilon_0}, \hat{n}$
- Formula: $E(r) = \begin{cases} 0 & (r < R) \\ \frac{q}{4\pi\epsilon_0 r^2}, \hat{n} & (r > R) \end{cases}$

Activities

Derive E for Wire, Plane, and Spherical Shell with Gauss's Law (10 min, experiment)

Materials: chalkboard or whiteboard, markers or chalk, meter stick, calculator for each pair, one printed A4 slip per pair with numerical example values (see teacher_instructions)

- Arrange students in pairs. Hand each pair a printed slip with one numerical example: (A) $\lambda = 5.0 \times 10^{-8} \text{ C/m}$ and $r = 0.20 \text{ m}$ for the wire; (B) $\sigma = 2.0 \times 10^{-6} \text{ C/m}^2$ for the infinite plane; (C) $q = 6.0 \times 10^{-6} \text{ C}$ and $R = 0.10 \text{ m}$ for the spherical shell and ask them to compute both outside ($r = 0.20 \text{ m}$) and inside ($r = 0.05 \text{ m}$). Say: 'You have 10 minutes to derive each formula from Gauss's law, show units at each step, and compute the field numerically for your given example.'
- Start a visible timer and announce time checkpoints at 6 minutes and 2 minutes remaining. While pairs work, circulate to check that each pair: (a) chooses an appropriate Gaussian surface, (b) writes Gauss's law in integral form closed integral $E \cdot dA = Q_{\text{enc}}/\epsilon_0$, and (c) carries units through each algebraic step.
- If a pair finishes early, direct them to the extension problem on their slip (derive the on-axis field for a finite uniformly charged disk at a point a distance z from the center) and provide

a scrap sheet for extra derivation. If a pair is stuck after 3 minutes, offer the support scaffold (below) and prompt them to start with the worked first item.

- After time is up, call for one pair to present the wire derivation, one pair the plane derivation, and one pair the spherical shell derivation; limit each presentation to 60 seconds and correct any missing unit or step on the board.

Student instructions:

- Work with your partner. Read your printed slip and identify which of the three cases (wire, plane, spherical shell) you will solve numerically and which you must still derive symbolically.
- Step 1 — Write Gauss's law in integral form: closed integral $E \cdot dA = Q_{enc}/\epsilon_0$. Explain aloud which Gaussian surface you choose and why its symmetry makes E constant over the surface portion you use. State the surface and its dimensions and label all distances with units (e.g., $r = 0.20$ m).
- Step 2 — Convert the surface integral into E times the appropriate area using symmetry. For each case show algebraic steps and include units: for the wire use a cylindrical surface of length L (m) and radius r (m) so closed integral $E \cdot dA = E(2\pi rL)$ with units $V/m \times m^2 = N \cdot m/C$? (keep SI units and show cancellation to get N/C); for the plane use a pillbox of area A (m^2) giving closed integral $E \cdot dA = E(2A)$ etc.; for the shell show that $Q_{enc} = q$ for $r > R$ and $Q_{enc} = 0$ for $r < R$ and use a spherical surface of area $4\pi r^2$ (m^2).
- Step 3 — Rearrange to solve for E symbolically, showing ϵ_0 with its units ($\epsilon_0 = 8.85 \times 10^{-12} C^2/(N \cdot m^2)$). Write the final formula with units: wire $E = \lambda/(2\pi \epsilon_0 r)$ (units N/C); plane $E = \sigma/(2\epsilon_0)$ (units N/C); shell outside $E = q/(4\pi \epsilon_0 r^2)$ (N/C) and shell inside $E = 0$ (N/C).
- Step 4 — Substitute the numerical values from your slip into the derived formula, carry units through the arithmetic, compute a numeric value and include units N/C . Record one clear sentence describing the physical meaning of the result (for example: 'At $r = 0.20$ m from the wire the magnitude of the electric field is ... N/C radially outward').

Success looks like: Each pair has drawn and labeled the Gaussian surface (cylinder, pillbox, or sphere) with distances in metres visible on their paper or board.; Each pair writes Gauss's law closed integral $E \cdot dA = Q_{enc}/\epsilon_0$ and shows the algebraic step where the surface integral becomes E times the area (e.g., $E \cdot 2\pi rL$ for the cylinder) while including units.; Each pair computes a numeric value with units (N/C) on their paper for their assigned numerical example and can point to the step where units cancel to give N/C .

Support: If you are stuck, start with this sentence frame and worked example for the wire: 'Choose a cylindrical Gaussian surface of radius $r = \underline{\hspace{1cm}}$ m and length $L = 1.0$ m. By symmetry E is constant on the curved surface and perpendicular to it. Gauss's law gives $E(2\pi rL) = Q_{enc}/\epsilon_0$. With linear charge density λ (C/m), $Q_{enc} = \lambda L$ (C). So $E = \lambda/(2\pi \epsilon_0 r)$. For $\lambda = 5.0 \times 10^{-8} C/m$ and $r = 0.20$ m: $E = (5.0 \times 10^{-8} C/m) / [2\pi(8.85 \times 10^{-12} C^2/(N \cdot m^2))(0.20 \text{ m})]$. Carry units: C/m divided by $(C^2/(N \cdot m^2)) \cdot m = N/C$. Compute with your calculator to get a numeric N/C value.'

Extension: Derive the electric field on the axis a distance z from the center of a finite uniformly charged disk of radius a and surface charge density σ by integrating concentric rings. Show units at every step and reduce your result to the plane limit ($a \rightarrow \text{infinity}$) to recover $E = \sigma/(2\epsilon_0)$.

Checkpoint Questions

- [analyze] You choose a cylindrical Gaussian surface of radius r and length L around an infinitely long straight wire with uniform linear charge density λ . Write the Gauss law integral and show how you get the magnitude of the electric field $E(r)$ at the curved surface. (Answer in symbols; state units for each quantity briefly.)

Expected: Symmetry: field is radial and constant on curved surface. Gaussian surface area = $(2\pi r L)$. Gauss's law: closed integral $E \cdot dA = E(2\pi r L) = Q_{\text{enc}}/\epsilon_0$. Enclosed charge $Q_{\text{enc}} = \lambda L$. So $E(2\pi r L) = \lambda L/\epsilon_0 \rightarrow E(r) = \lambda/(2\pi \epsilon_0 r)$. Units: λ in C/m, ϵ_0 in $C^2/(N \cdot m^2)$, r in m $\rightarrow E$ in N/C (or V/m).

- [analyze] Sketch a Gaussian pillbox that straddles an infinite charged plane with surface charge density σ . Use Gauss's law to find the magnitude of the electric field on one side of the plane. Explain why the field does not depend on distance from the plane.

Expected: Gaussian pillbox has two flat faces area A , one on each side of plane. Field normal to plane and constant on faces. Flux = $E \cdot A + E \cdot A = 2EA$. Enclosed charge = σA . Gauss: $2EA = \sigma A/\epsilon_0 \rightarrow E = \sigma/(2\epsilon_0)$. Field is constant with distance because plane is infinite: symmetry means no preferred distance and field lines are straight and normal, so E doesn't vary with r .

- [analyze] For a thin spherical shell of radius R carrying total charge q , pick Gaussian surfaces for $r < R$ and $r > R$ and state the enclosed charge in each case. From that, write $E(r)$ for both regions and state the physical units.

Expected: For $r < R$: choose spherical Gaussian surface radius r . Enclosed charge = 0 (shell outside the surface) $\rightarrow$ closed integral $E \cdot dA = 0 \rightarrow E = 0$ (N/C). For $r > R$: choose radius r . Enclosed charge = $q \rightarrow E(4\pi r^2) = q/\epsilon_0 \rightarrow E(r) = q/(4\pi \epsilon_0 r^2)$ radial outward. Units: q in C, r in m, ϵ_0 in $C^2/(N \cdot m^2) \rightarrow E$ in N/C.

EXIT TICKET

A long charged wire has $\lambda = 5.0 \times 10^{-8}$ C/m. A parallel infinite charged plane has $\sigma = 8.85 \times 10^{-12}$ C/m². A thin spherical shell of radius 0.50 m carries total charge $q = 2.0 \times 10^{-6}$ C. For each object, write the expression for the electric field magnitude at the following points and compute its numerical value (show units): (a) at $r = 0.10$ m from the wire axis, (b) at 0.20 m on one side of the plane, (c) at $r = 0.25$ m (inside) and $r = 1.0$ m (outside) the spherical shell.

Success indicator: Student submits a short answer with four items: (a) $E_{\text{wire}} = \lambda/(2\pi \epsilon_0 r)$ and numerical value $\sim 5.0e-8/(2\pi \cdot 8.85e-12 \cdot 0.10) \sim 9.0 \times 10^3$ N/C (radial); (b) $E_{\text{plane}} = \sigma/(2\epsilon_0)$ and numerical value $\sim 8.85e-12/(2 \cdot 8.85e-12) = 0.5$ N/C (normal); (c1) $E_{\text{inside_shell}} (r=0.25 \text{ m}) = 0$ N/C; (c2) $E_{\text{outside_shell}} (r=1.0 \text{ m}) = q/(4\pi \epsilon_0 r^2)$ and numerical value $\sim 2.0e-6/(4\pi \cdot 8.85e-12 \cdot 1.0^2) \sim 1.8 \times 10^5$ N/C, with units N/C. (Numbers may vary slightly with rounding.)

Homework

- Derive the electric field of an infinite line charge using Gauss's law, but this time show every algebra step for solving $E(r)$ from $E(2\pi r L) = \lambda L/\epsilon_0$; write units at each step. (10 minutes)
- A pair-problem: Two infinite parallel planes carry equal and opposite surface charge densities σ and $-\sigma$. Sketch field lines and use superposition of the single-plane result to write E in the three regions (left of both, between, right of both). Give numerical answers for $\sigma = 4.0 \times 10^{-9}$ C/m². (15 minutes)
- Calculate the work (in joules) required to bring a $1.0 \mu\text{C}$ test charge from $r = 2.0$ m to $r = 0.75$ m outside the charged spherical shell ($q = 2.0 \times 10^{-6}$ C, $R = 0.5$ m). Use the shell's outside-field expression. Show units in each line. (20 minutes)

Estimated time: 45 min

MENTOR MOMENT — SEEING SYMMETRY SAVE TIME

A student once tried to integrate Coulomb's law over a long wire point-by-point for hours; after we recognized cylindrical symmetry and used Gauss's law, the answer took five minutes.

Takeaway: Before you start heavy integration, ask: what symmetry can let Gauss's law or simple geometry do

the work?

Learning Gaps & Misconceptions

[high] When you rub two objects together you make new electric charge — rubbing creates charge out of nothing (for example, rubbing a balloon on your hair makes charge appear).

- **Diagnostic:** You rub a balloon on your hair. Afterwards the balloon sticks to the wall and your hair looks different. Consider the isolated system consisting of the balloon plus your hair (no charges can leave this pair). Is the total electric charge of that isolated system after rubbing greater than, less than, or the same as it was before rubbing? Explain numerically if possible.

Reveals: Saying the total charge is greater (or less) shows the student believes rubbing creates (or destroys) charge rather than transferring it; they haven't applied conservation of charge to the two-object system.

- **Diagnostic:** Two small identical metal pith balls are electrically neutral. You rub one ball with a cloth, then bring the rubbed ball into contact with the second ball and then separate them. After the whole process, what is the net charge of the two-ball system compared with the start? Give an answer in coulombs or in words.

Reveals: Answering that the net charge is nonzero indicates the student thinks rubbing (or the rubbing+contact process) created net charge, rather than redistributed existing electrons between the two balls or between ball and cloth.

- **Remediation:** In-class quantitative demonstration: Place two conductors (A and B) connected together and measure the total charge using an electrometer (show reading in coulombs). Disconnect them, rub A with a cloth, then immediately measure the charge on A and on B separately and then measure the sum $Q_A + Q_B$. Display the numeric readings and units (C). Repeat once with the apparatus isolated (insulated stand) so charge cannot escape the pair.

Why: A direct measurement in coulombs with the pair isolated shows empirically that $Q_A + Q_B$ equals the original total (within measurement error). This confronts the idea that rubbing 'creates' charge by anchoring charge to a measurable quantity with units and demonstrating conservation in the actual system the student is considering.

- **Remediation:** Microscopic accounting exercise: show a diagram of the relevant atoms for the two bodies and the cloth. Model charge as integer electron counts: write $q = n \cdot e$ for each body, with $e = 1.602 \times 10^{-19} \text{ C}$ explicitly stated and carried through a short calculation. Have students calculate how many electrons must move to produce a measured charge (e.g. $1.6 \times 10^{-8} \text{ C}$ corresponds to 1.0×10^{11} electrons transferred).

Why: Connecting macroscopic charge measurements to electron counts and the formula $q = n \cdot e$ explains that charge changes come from moving integer electrons between bodies, not creating charge. Seeing the huge number of electrons involved prevents the intuitive leap that a small rubbing action 'creates' something mysteriously new, and the numerical derivation reinforces conservation by showing equal and opposite electron count changes.

- **Remediation:** Structured prediction and explanation: present a brief thought experiment and worksheet. Ask students to predict the net charge of an insulated box containing two objects before and after rubbing them together (no contact with outside). Then guide a class discussion: ask where any extra charge would come from and force them to name the location (other object, cloth, Earth). If any student predicts net charge creation, have them

reconcile that claim with the electron-count model and the electrometer data from the demo.

Why: By forcing explicit predictions about an isolated system and then requiring reconciliation with the microscopic/electrometer evidence, students must revise the mental model that rubbing creates charge. The step-by-step reasoning (isolate system, identify carriers, account electrons) trains them to apply conservation rather than relying on ambiguous everyday phrasing.

[high] If the total electric flux through a closed surface is zero, there is no charge inside that surface.

- Diagnostic: A small $+1\ \mu\text{C}$ charge and a small $-1\ \mu\text{C}$ charge are both placed inside a closed spherical surface. You measure the total electric flux through that sphere and find it is zero. Does this measurement prove there are no charges inside the sphere? Explain.

Reveals: Whether the student thinks zero net flux implies no charges are present (confuses net flux with presence of charge), rather than recognising net enclosed charge could be zero even when charges exist.

- Diagnostic: An electric field is zero everywhere on the surface of a closed box, but measurements inside the box show a nonzero field near some locations and a charge sitting at the centre. From Gauss's law, is the net flux through the box necessarily zero? If the flux is zero, does that mean there are no charges inside?

Reveals: Whether the student realises that flux is the surface integral of $\mathbf{E} \cdot d\mathbf{A}$ (not simply 'field zero at the surface' in general), and that zero net flux can coexist with nonzero internal charge distributions that sum to zero net enclosed charge.

- Remediation: Start with a concrete drawing/animation of field lines for three cases: (A) a single $+Q$ at the centre of a sphere, (B) a $+Q$ and $-Q$ pair (dipole) fully inside the sphere, and (C) two $+Q$ charges inside. For each case have students count (or estimate) the number of field lines leaving minus entering the surface and compute net flux = (number of outgoing lines – incoming lines).

Why: Visualising and counting lines ties the abstract flux integral to a measurable bookkeeping activity. It shows that case B can have equal numbers of lines leaving and entering so the algebraic sum (net flux) is zero even though charges exist inside, correcting the 'zero flux $\Rightarrow$ no charge' inference.

- Remediation: Give students a short numeric activity: compute (or use simple code/spreadsheet) the surface integral closed integral $\mathbf{E} \cdot d\mathbf{A}$ for (i) a point charge at centre of a sphere, (ii) a dipole inside a sphere, and (iii) two equal charges of opposite sign. Provide explicit formulas for $\mathbf{E}$ from each charge and discrete patches on the sphere so they sum contributions and observe the net result equals $Q_{\text{enclosed}}/\epsilon_0$.

Why: Carrying units and sums step by step ties the symbolic Gauss's-law formula to measured contributions. The computation exposes that contributions from individual charges add algebraically and can cancel, so zero net flux corresponds to zero net enclosed charge, not to absence of charge.

- Remediation: Lead a guided questioning sequence: present a closed surface with measured zero net flux and ask 'What can you conclude about the total charge enclosed? What can you conclude about whether any charge exists at all inside? Where would you look next to decide?' Encourage phrasing answers in terms of $Q_{\text{enclosed}} = \epsilon_0 \cdot \Phi$ and distinguishing 'net charge' from 'presence of charges'.

Why: Socratic prompts force students to state the precise logical implication of Gauss's law (zero flux $\Rightarrow$ zero net enclosed charge) and the limits of that conclusion (it does not imply absence of individual charges). Requiring explicit unit-based statements (flux in Nm^2/C , charge in C) reinforces correct application.

[high] I can just use $F = k q_1 q_2 / r^2 \hat{r}$ without worrying which way $\hat{r}$ points — the sign of the product $q_1 q_2$ tells me attraction or repulsion and fixes the direction, so I don't need to define the unit vector from one charge to the other carefully.

- Diagnostic: Charges $q_1 = +2.0 \mu\text{C}$ at $x = 0.00 \text{ m}$ and $q_2 = -3.0 \mu\text{C}$ at $x = +0.50 \text{ m}$. Using the vector form $F = k q_1 q_2 / |r_2 - r_1|^2 \hat{r}$, with $\hat{r}$ defined as $(r_2 - r_1)/|r_2 - r_1|$, write the force vector on q_1 due to q_2 (give magnitude, sign/direction along x , and units).

Reveals: Whether the student correctly computes the displacement $r_2 - r_1$ and uses $\hat{r}$ in the ordered sense (force on 1 due to 2 uses $r_2 - r_1$). A student who ignores the $\hat{r}$ definition will likely give a direction reversed or will try to 'fix' direction using only the sign of $q_1 q_2$.

- Diagnostic: Two positive charges $q_A = +4 \mu\text{C}$ at $x = 0 \text{ m}$ and $q_B = +1 \mu\text{C}$ at $x = +0.20 \text{ m}$. Without calculating numbers, state the direction of the force on q_B due to q_A and then write the vector expression using the displacement r_A and r_B (explicitly show $r = r_B - r_A$ or $r_A - r_B$).

Reveals: Whether the student knows which displacement corresponds to the force on the second charge (must use $r_B - r_A$ when writing $F_{\text{on}_B \text{ due to } A}$) rather than assuming a generic ' $\hat{r}$ from A to B' regardless of order.

- Remediation: Demonstration + lab: set two small charged spheres (or pith balls) on insulating stands at labelled positions $x = 0.00 \text{ m}$ and $x = +0.40 \text{ m}$. Mark the position vectors r_1 and r_2 on a whiteboard and draw arrows showing $r_2 - r_1$ and $r_1 - r_2$. Ask students to predict and then measure the direction of the force on each ball, then draw the force vectors on the diagram with units (N).

Why: Concrete physical forces plus labelled position vectors ties the abstract symbol $\hat{r}$ to a measurable direction. Seeing both forces simultaneously emphasises that the same pair of charges produces equal-and-opposite vectors and that which displacement you form ($r_2 - r_1$ vs $r_1 - r_2$) determines the arrow direction for the specific ordered force.

- Remediation: Derivation walk-through: on the board, start from the displacement vector $r = r_1 - r_2$ (explicit coordinates) and derive $F_{12} = k q_1 q_2 (r_1 - r_2) / |r_1 - r_2|^3$ step by step. Use a numeric example (plug numbers and units) so students compute both magnitude and the three-component vector. Emphasise the subscript notation (force on 1 due to 2) at every step and keep SI units in every line.

Why: Showing how the algebra forces a particular vector direction addresses the root error (treating direction as an afterthought). Carrying units and a numeric example prevents hiding sign/direction errors in abstract symbols and forces students to compute the direction from the displacement rather than guessing from verbal 'attract/repel'.

- Remediation: Quick practice: give pairs of short coordinate problems (3–4) where students must write the vector expression for $F_{\text{on}_A \text{ due to } B}$ and $F_{\text{on}_B \text{ due to } A}$, explicitly showing the displacement used (e.g. $r_A = (0,0,0)\text{m}$, $r_B = (0.3,0,0)\text{m}$; $q_A = -2 \mu\text{C}$, $q_B = +5 \mu\text{C}$). Students swap answers to check each other's ordered $\hat{r}$ and sign.

Why: Repeated, low-pressure practice with different sign combinations and coordinates makes forming the ordered displacement routine. Peer checking catches the common swap-of-order error and reinforces that the sign of $q_1 q_2$ gives attraction/repulsion but the actual arrow comes from the chosen displacement vector.

[medium] Only conductors can become electrically charged; insulators cannot acquire charge.

- **Diagnostic:** You rub a plastic comb on a dry wool sweater and then pick up small bits of paper with the comb. Is the comb electrically charged after rubbing? Explain your answer.

Reveals: A student who says 'no' or claims the comb can't be charged is holding the misconception that insulators cannot acquire charge; they are equating lack of conduction with inability to hold charge.

- **Diagnostic:** You have two objects: a metal sphere on an insulating stand and a plastic ruler. You rub both with wool. After rubbing, which object will keep its charge longer when left isolated on the lab bench? Explain why.

Reveals: If the student answers that only the metal can be charged or that the plastic can't hold charge, they believe insulators cannot acquire/retain charge and confuse leakage to ground with the ability to be charged.

- **Remediation: Demonstration:** Rub a nylon/acetate rod or plastic comb with wool and use an electroscope to show the electroscope leaves diverge — record the angle of divergence. Then touch the electroscope to ground and repeat the rubbing so students see the divergence disappear. Next, do the same rubbing with a metal sphere on an insulating stand; show directly that the metal sphere when charged and then connected to ground loses its charge quickly while the plastic piece still causes paper bits to be attracted and the electroscope reacts differently.

Why: The paired observations confront the incorrect generalisation: students see an insulator (plastic) clearly produce measurable electrostatic effects (electroscope deflection, attraction of paper) even though charges don't flow to ground. Showing the metal losing charge quickly when grounded isolates the separate ideas 'can be charged' and 'allows charges to move to ground.' Using the electroscope gives a quantitative, repeatable measure (angle or deflection) to anchor the abstract notion of 'amount of charge' to a measurable quantity.

- **Remediation: Guided experiment:** Have pairs measure how long charge persists. Charge a plastic rod by rubbing, bring it near small paper pieces and an electroscope and time how long attraction/deflection persists. Repeat with a metal object charged (using induction and then isolating it on an insulating stand). Record times and environmental conditions (humidity, distance). Students plot persistence (s) vs. object type and discuss differences.

Why: Measuring charge persistence ties the abstract claim 'insulators can't be charged' to concrete data: insulators often retain localized charge longer because charges cannot move to ground, whereas conductors allow redistribution and loss. Recording units (seconds) and conditions prevents hand-waving and addresses the mistaken inference from 'doesn't conduct' to 'can't be charged.' The student-generated data undermines the misconception by induction from controlled observations.

- **Remediation: Directed question sequence and derivation:** Lead students through a short reasoning sequence on the whiteboard with units: (1) define charge transfer (electrons moved between surfaces) with a simple model (count of electrons), (2) show how rubbing transfers electrons from material A to B using atomic-level sketches, and (3) derive why an insulator holds localized excess charge (charges trapped, no path to ground) while a conductor redistributes charge and can flow to ground if connected — include example numbers (e.g. $1\text{e-9 C} \sim 6.25 \times 10^9$ electrons) and predict electroscope deflection using proportional reasoning.

Why: Deriving the result step-by-step and anchoring with explicit units and approximate numbers confronts the underlying faulty generalisation. Students see the mechanism (electron transfer) that makes charging possible for any material and the separate role of mobility. Providing numeric scale links the abstract idea of 'some charge' to measurable quantities and explains why insulators visibly hold charge even though they are

non-conducting.

[medium] The symbol for charge density (ρ , σ or λ) is just the amount of charge in a little piece by itself, so when I set up integrals I can write $dq = \rho$ (or $dq = \lambda(x)$) and don't need to multiply by dV , dA or dx — the density already IS that small charge.

- Diagnostic: You have a thin rod of length L with linear charge density that varies as $\lambda(x) = \alpha x$ (where x is measured from one end). To set up the integral for the total charge Q on the rod, which expression for the infinitesimal charge do you use?

Reveals: Whether the student recognises density as charge per unit length and correctly includes the differential length element.

- Diagnostic: What are the SI units of volume charge density ρ ? (Give the units, not a numerical value.)

Reveals: Whether the student understands density as charge per unit volume rather than just 'charge'.

- Remediation: Start with a concrete discrete model: lay out 4 identical small plastic beads each labelled q on a metre stick segment. Ask students to write the total charge as $Q = q_1 + q_2 + q_3 + q_4$, then group the stick into N equal segments and express charge as a sum $\sum \lambda_i \Delta x_i$ (where $\lambda_i = q_i / \Delta x_i$). Then physically reduce Δx (more beads, smaller pieces) and take the limit to show $dq = \lambda(x) dx$ and $Q = \int \lambda(x) dx$.

Why: This connects the continuous formula to the discrete sum they already accept. Showing the limiting process makes clear that density is a per-length quantity that must be multiplied by a length to get charge (it addresses the root confusion about 'what the symbol represents').

- Remediation: Run a units-check exercise: give several expressions used in integrals (e.g. $\lambda(x) dx$, ρdV , σdA , $\lambda(x)$, ρ , dq) and ask students in pairs to write the SI units for each and say whether each is a charge or not. Require them to correct a deliberately incorrect line (e.g. 'dq has units $C \cdot m^{-3}$ ').

Why: Forcing explicit unit tracking exposes the per-unit nature of densities; when students see that λ has units $C \cdot m^{-1}$ while $\lambda \cdot dx$ has C , they shift from treating symbols as standalone charges to seeing the multiplication by a measure as necessary to get charge.

- Remediation: Give a short scaffolded problem: a nonuniform rod with $\lambda(x) = kx^2$. Ask first for an expression for the charge on a small piece from x to $x + \Delta x$ (expect $\lambda(x)\Delta x$), then let $\Delta x \rightarrow 0$ to write dq , then integrate. Follow with a conceptual question: 'If λ has units $C \cdot m^{-1}$, could dq equal λ without a dx ? Explain.'

Why: A scaffolded sequence moves students from an intuitive finite piece to the differential expression and then forces a verbal justification connecting units, algebra and the limiting process — reinforcing the correct mental model rather than merely stating it.

[medium] If I double the distance between two charges, the force becomes half as big (and if I halve the distance the force doubles).

- Diagnostic: Two point charges are 2.0 cm apart and repel each other with force F . If the separation is increased to 4.0 cm, what is the new force in terms of F ?

Reveals: Whether the student recognises the inverse-square dependence rather than treating distance changes as linear.

- Diagnostic: Two identical charges at distance r exert force F . If you bring them to distance $r/3$, what is the new force in terms of F ?

Reveals: Whether the student correctly applies the squared factor when distance decreases (large changes give clear numerical contrast).

- **Remediation:** Give students three concrete numerical examples on the board. Use identical charges $q_1 = q_2 = 1.0 \mu\text{C}$ and $k = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Compute F at $r = 1.0 \text{ m}$, $r = 2.0 \text{ m}$ and $r = 0.5 \text{ m}$ step by step, showing units: $F_1 = kq^2/1^2$, $F_2 = kq^2/2^2$, $F_3 = kq^2/(0.5)^2$. Have students calculate the ratios F_2/F_1 and F_3/F_1 .

Why: Concrete numbers and carried units force students to manipulate the algebra and see actual numerical factors ($F_2 = 1/4 F_1$, $F_3 = 4 F_1$). This confronts the incorrect 'half' or 'triple' intuition with unambiguous arithmetic results rather than abstract claims.

- **Remediation:** Use a computer simulation or video of a Coulomb-force experiment (or a demonstrator with charged pith balls if available). Have students collect or observe measured force values at several distances. Then guide them to plot force F versus distance r (linear axes) and separately plot F versus $1/r^2$ (or plot F on vertical axis and $1/r^2$ on horizontal axis).

Why: Seeing raw data and comparing the two plots makes the difference visible: F vs r is non-linear, while F vs $1/r^2$ is approximately linear through the origin. This addresses the root cause (confusing inverse with inverse-square) by giving visual, empirical evidence of the squared relationship.

- **Remediation:** Lead a short derivation/justification: start from the fitted linear relation $F = k q_1 q_2 (1/r^2)$ found in the plotted data; then discuss the geometric reason (flux through a sphere grows with r^2) qualitatively, and carry units through to show k has units $\text{N}\cdot\text{m}^2/\text{C}^2$ so F must scale as $1/r^2$. Finish with quick mental-check problems (e.g. 'If r increases by factor 5, what happens to F ?) that require using the squared factor.

Why: Connecting the algebraic formula, units and a geometric explanation (surface area of a sphere proportional to r^2) ties the inverse-square law to physical intuition and unit analysis. This corrects the misconception by replacing informal linear intuition with linked mathematical, empirical and geometric reasoning.